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Faculty of Engineering, Built Environment and Information Technology

ELI 220

Practical Report 2

Group 16

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October 2, 2020

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A Python Code **45**

1 Introduction

The Fourier transform is used to approximate a pulse train and graphically plot this in the frequency domain. "LTspice" is used to simulate the actual components and give a graphical representation of the expected outputs. Python is used to graphically plot the different Fourier- and Laplace transforms.

2 Preparatory Work

2.1 Question 1

$$
\begin{aligned}
c(t) &= rect\left(\frac{t}{\tau}\right) \times \sum_{n=-\infty}^{\infty} \delta\left(t - nT\right) \\
c(t) &= \int_{-\infty}^{\infty} rect\left(\frac{t}{\tau}\right) e^{-j\omega t}\ dt \\
&= \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} (1)\, e^{-j\omega t}\ dt \\
&= -\frac{1}{j\omega}\left(e^{-j\omega t}\right)|_{\frac{\tau}{2}}^{\frac{\tau}{2}} \\
&= -\frac{1}{j\omega}\left(e^{-j\omega\frac{\tau}{2}} - e^{j\omega\frac{\tau}{2}}\right) \\
\text{thus } c(\omega) &= \frac{2sin\left(\frac{\omega\tau}{2}\right)}{\omega} \\
&= \tau\frac{sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}} \\
&= \tau sinc\left(\frac{\omega\tau}{2}\right)
\end{aligned}
$$

$$
\begin{aligned}
\sum_{n=-\infty}^{\infty} \delta\left(t - nT\right) = c(\omega) &= \omega_0 \sum_{n=-\infty}^{\infty} \delta\left(\omega - n\omega_0\right) \\
\omega_0 &= \frac{2\pi}{T} \\
c(\omega) &= \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n\frac{2\pi}{T}\right) \\
\text{thus } c(f) &= \left(\frac{2\pi f\tau}{2}\right) \times \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n\frac{2\pi}{T}\right)
\end{aligned}
$$

a)

$$
\begin{aligned}
-20 \leq & f \leq 20 \text{ Hz} \\
\tau &= \frac{T}{2} \\
T &= 1
\end{aligned}
$$

$$c(f) = \frac{1}{2} sinc\left(\frac{2\pi f \frac{1}{2}}{2}\right) \times 2\pi \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n2\pi\right)$$
$$= \frac{1}{2} sinc\left(\frac{\pi f}{2}\right) \times 2\pi \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n2\pi\right)$$

Unit impulse (δ) will be 1 for all $n = f$

$$c(f) = \frac{1}{2} sinc\left(\frac{\pi f}{2}\right) \times 2\pi$$
$$c(f) = \pi sinc\left(\frac{\pi f}{2}\right)$$

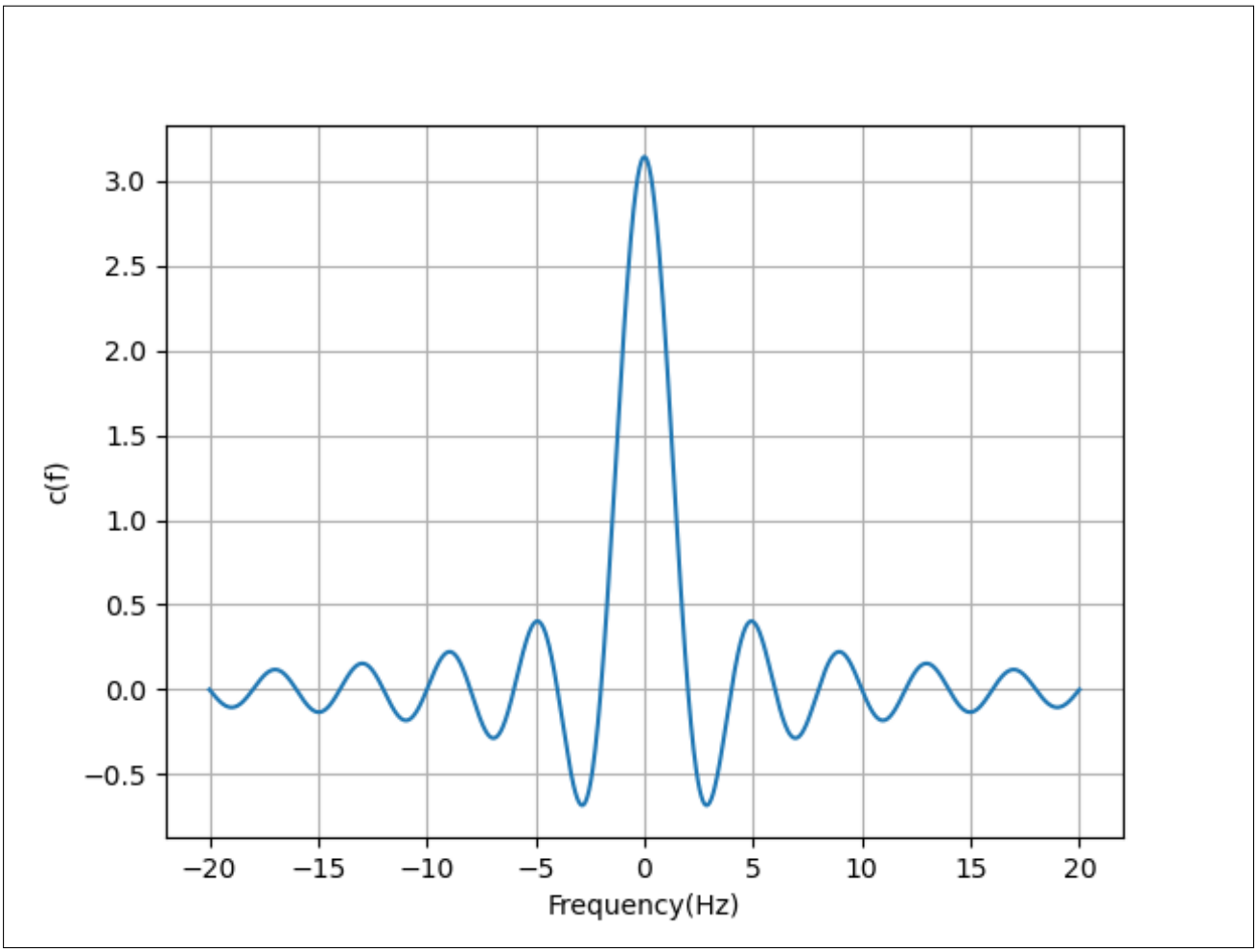

Figure 2.1.1: Plotting of a

b)

$$-40 \leq f \leq 40 \text{ Hz}$$
$$\tau = \frac{T}{20}$$
$$T = 1$$

$$
\begin{aligned}
c(f) &= \frac{1}{20} sinc\left(\frac{2\pi f \frac{1}{20}}{2}\right) \times 2\pi \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n2\pi\right) \\
&= \frac{1}{20} sinc\left(\frac{\frac{1}{10}\pi f}{2}\right) \times 2\pi \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n2\pi\right)
\end{aligned}
$$

Unit impulse (δ) will be 1 for all $n = f$

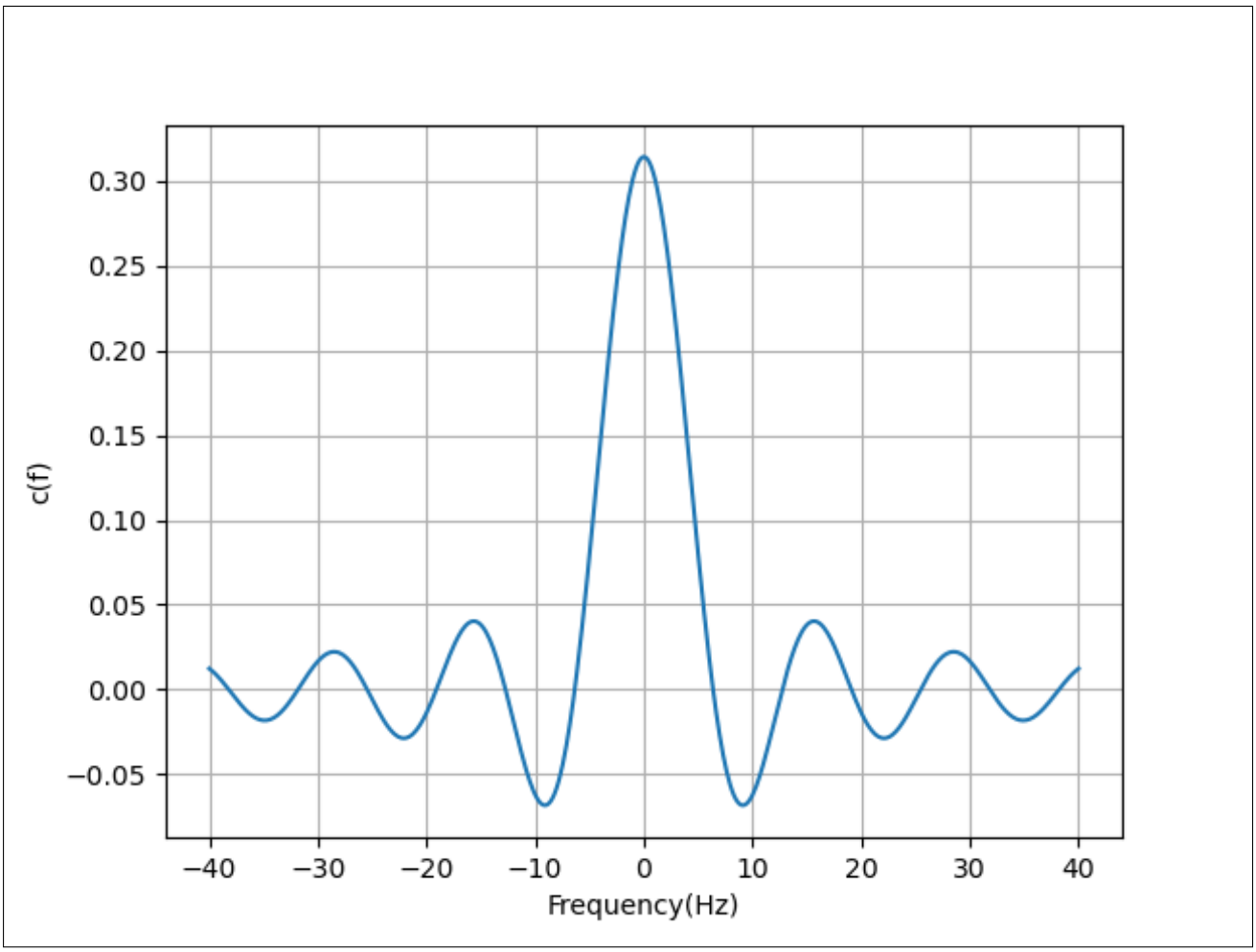

Figure 2.1.2: Plotting of b

c)

$$
\begin{gathered}
-40 \leq f \leq 40 \text{ Hz} \\
\tau \to 0 \\
T = 1
\end{gathered}
$$

$$
\begin{aligned}
c(f) &= \tau \left(\frac{sin\left(\frac{2\pi f\tau}{2}\right)}{\frac{2\pi f\tau}{2}} \right) \times 2\pi \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n2\pi\right) \\
&\lim_{\tau\to 0} \left(\tau \frac{sin\left(\frac{2\pi f\tau}{2}\right)}{\frac{2\pi f\tau}{2}} \right) \\
&= \lim_{\tau\to 0}(\tau) \times \lim_{\tau\to 0} \left(\frac{sin\left(\frac{2\pi f\tau}{2}\right)}{\frac{2\pi f\tau}{2}} \right) \\
&= 0 \times \pi \\
&= 0 \\
\text{thus } c(f) &= 0 \times 2\pi \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - n2\pi\right) \\
&= 0
\end{aligned}
$$

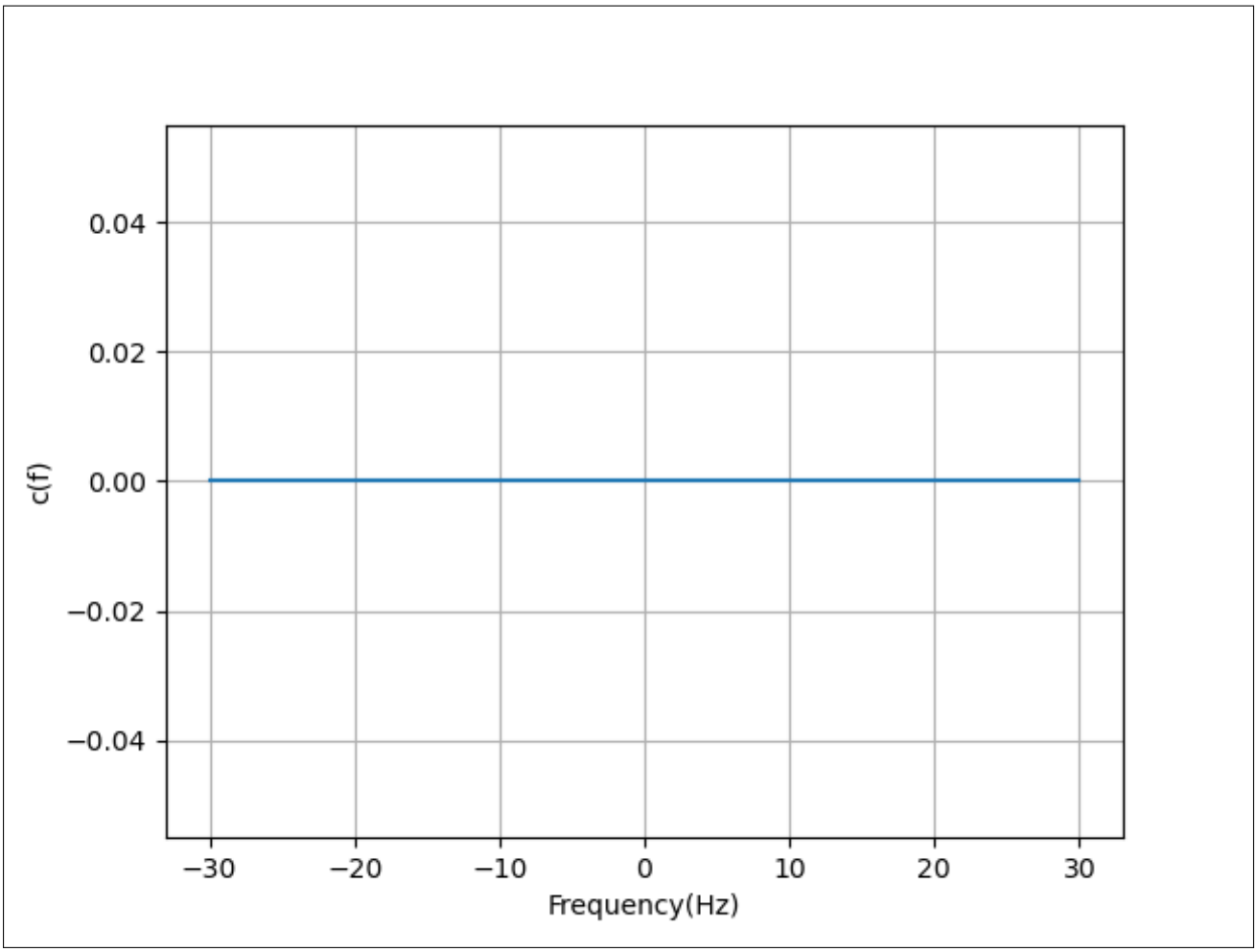

Figure 2.1.3: Plotting of c

d)

$$-30 \leq f \leq 30 \text{ kHz}$$
$$\tau = 75 \times 10^{-6} s$$
$$T = 3.7 \times 10^{-3} s$$

$$c(f) = \tau sinc\left(\frac{2\pi f \tau}{2}\right) \times \frac{2\pi}{3.7 \times 10^{-3}} \sum_{n=-\infty}^{\infty} \delta\left(2\pi f - \frac{2\pi}{3.7 \times 10^{-3}} n\right)$$

for some value of n in the impulse function (δ) the awnser will be 1, this is for each change in f between $-30 \leq f \leq 30$ kHz

$$\text{Thus } c(f) = 75 \times 10^{-6} sinc\left(\frac{2\pi f \times 75 \times 10^{-6}}{2}\right) \times \frac{2\pi}{3.7 \times 10^{-3}}$$

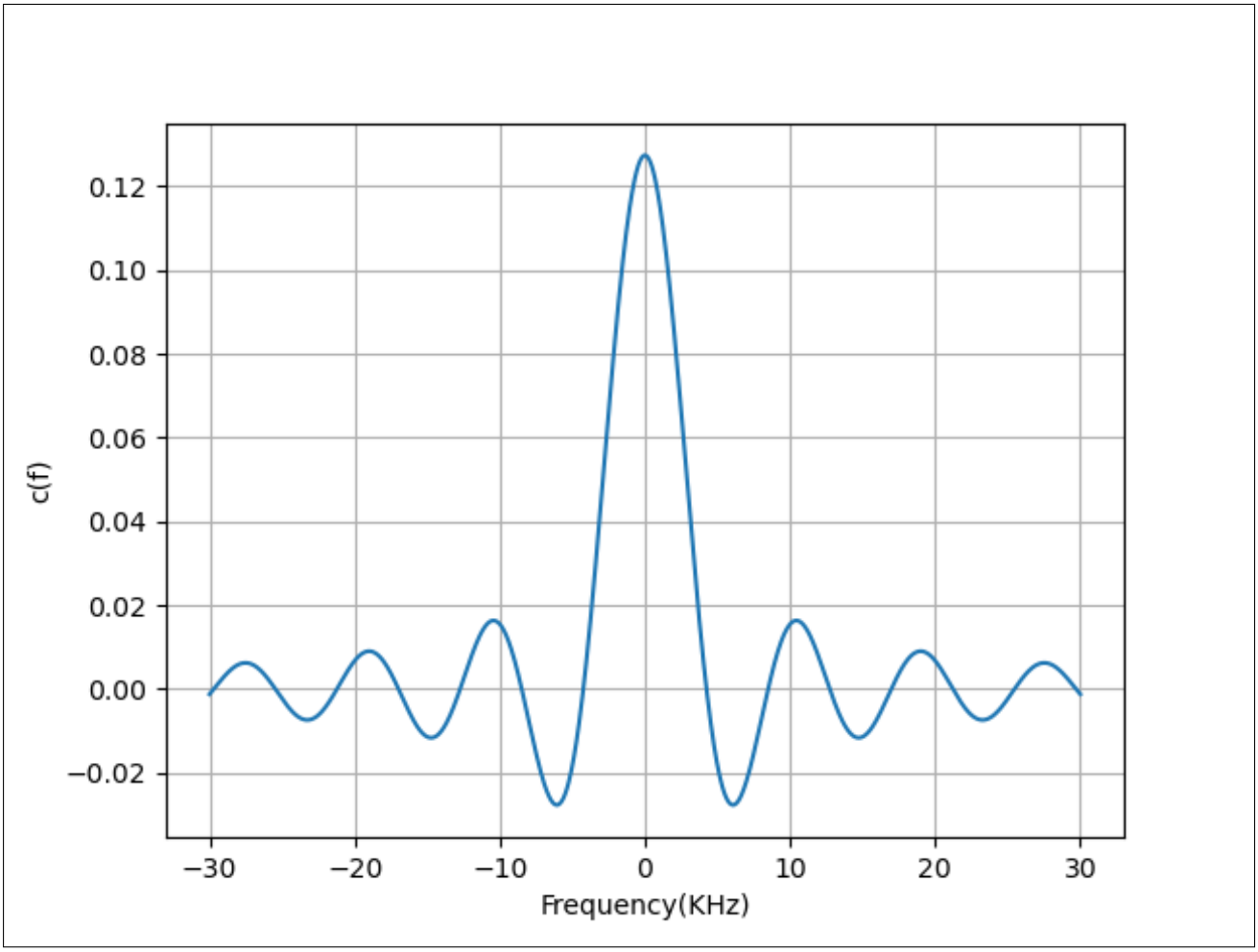

Figure 2.1.4: Plotting of d

2.2 Question 2

a) i)

$$|H(f)| \rightarrow \text{1Hz to 1mHz}$$
$$h(t) \rightarrow \text{0 to 1.5ms}$$

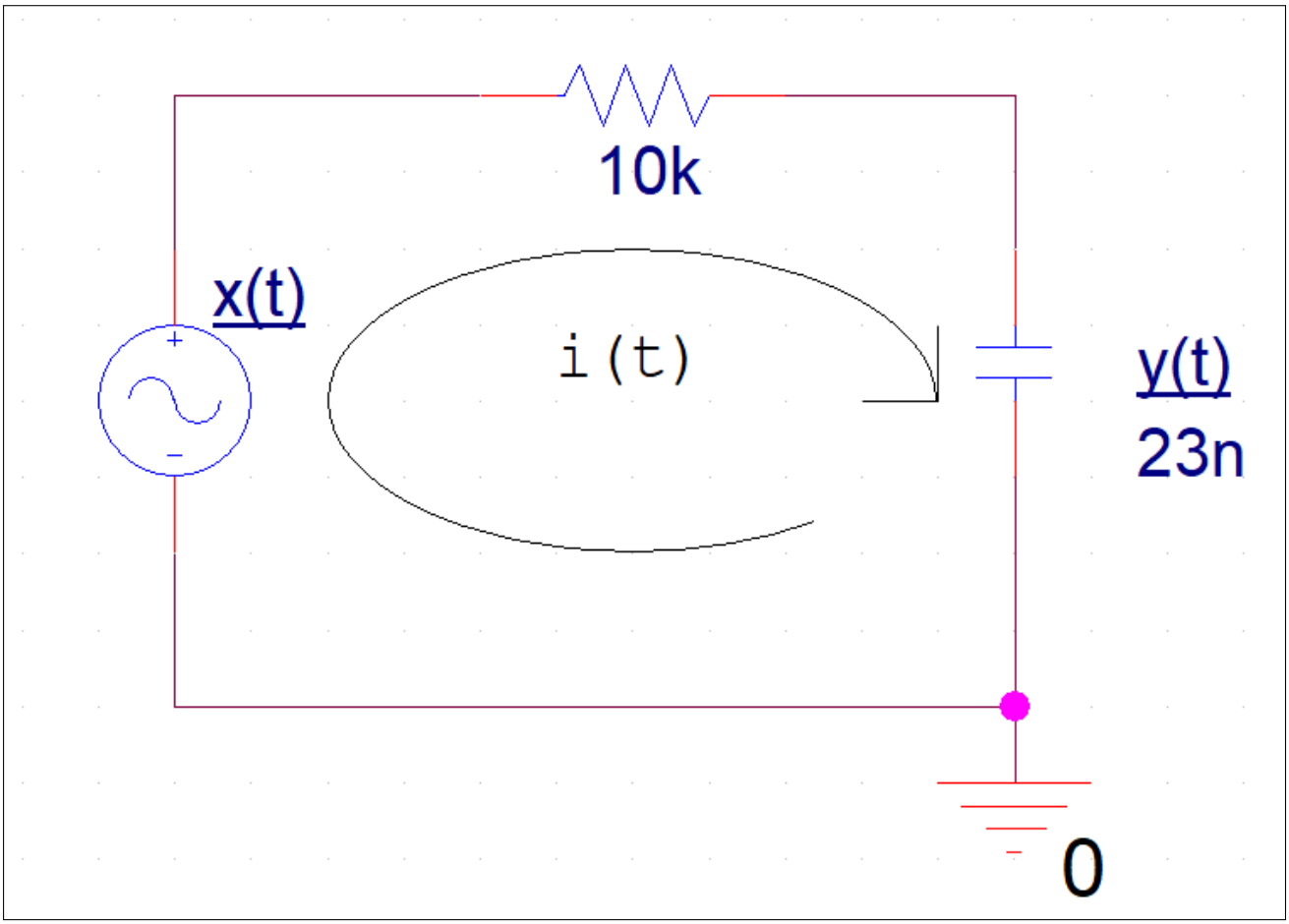

Figure 2.2.1: Circuit used in (i) calculations

$$\begin{aligned} i(t) &= c\frac{d}{dt} \times y(t) \\ &= 33 \times 10^{-9} D \times y(t) \end{aligned}$$

Use KVL:

$$\begin{aligned} x(t) &= 1000i(t) + y(t) \\ x(t) &= 1000\left(33 \times 10^{-9} D \times y(t)\right) + y(t) \\ x(t) &= 3.3 \times 10^{-4} D y(t) + y(t) \\ x(t) &= y(t)(3.3 \times 10^{-4} D + 1) \end{aligned}$$

$$\text{Thus } P(D) = 1$$
$$3.3 \times 10^{-4} D + 1 = 0$$
$$\text{Thus } 3.3 \times 10^{-4} \lambda + 1 = 0$$

$$\lambda = \frac{-1}{3.3 \times 10^{-4}}$$
$$\lambda = -3030.30$$

$$\text{Thus } y_n = C_1 e^{\lambda t}$$
$$\lambda = C_1 e^{-3030.30t}$$

It is a first order circuit because $N = 1$.

$$y_n^{(1-1)}(0) = 1 = C_1 e^{-3030.30(0)}$$
$$1 = C_1 e^0$$
$$C_1 = 1$$

$$\therefore y_n(t) = e^{-3030.30(t)}$$
$$h(t) = (P(D) y_n(t))\, u(t)$$
$$\therefore h(t) = e^{-3030.30(t)} \times u(t)$$

Transform $h(t)$:

$$h(t) = e^{-3030.30(t)} \times u(t)$$
$$H(s) = \int_0^\infty e^{-3030.30t - st}\, dt$$
$$= \int_0^\infty e^{-(s+3030.30)t}\, dt$$
$$= -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^\infty$$

According to the Laplace table:

$$e^{-3030.30(t)} \times u(t) \rightleftharpoons \frac{1}{s + 3030.30}$$
$$\therefore H(\omega) = \frac{1}{j\omega + 3030.30}$$
$$H(f) = \frac{1}{j2\pi f + 3030.30}$$
$$|H(f)| = \frac{1}{\sqrt{(2\pi f)^2 + (3030.30)^2}}$$

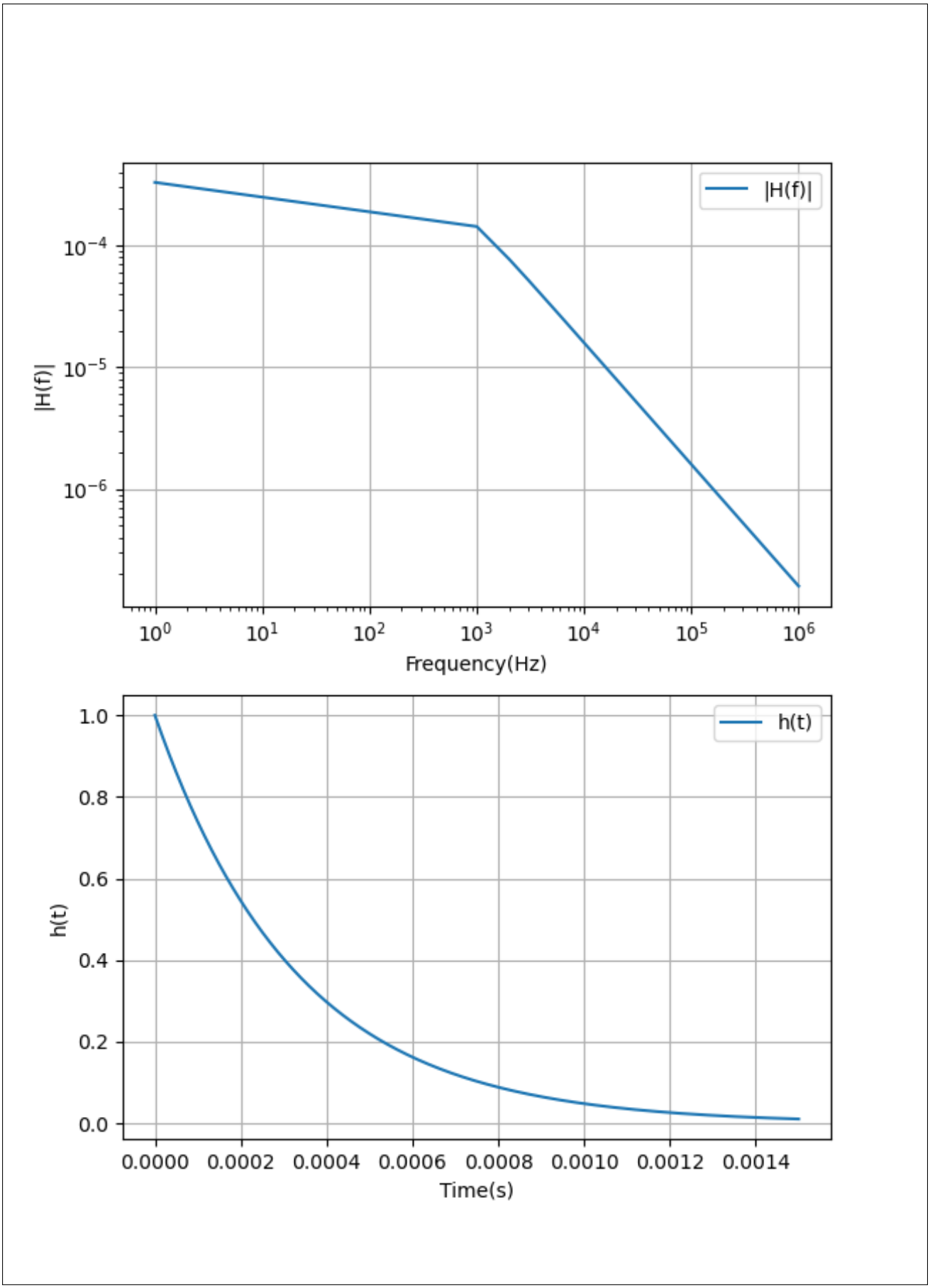

Figure 2.2.2: Plotting of i

ii)

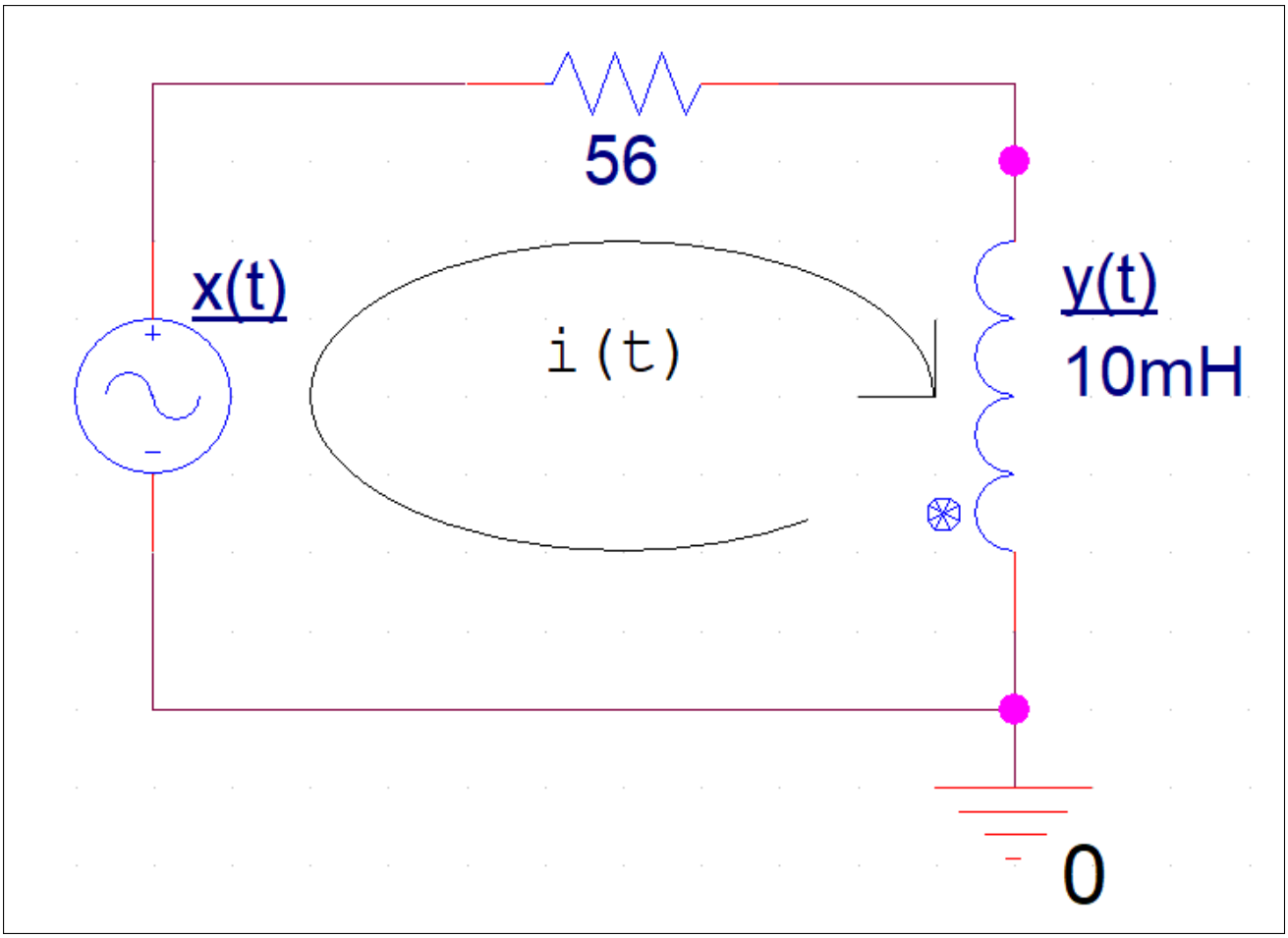

Figure 2.2.3: Circuit used in (ii) calculations

Using KVL:

$$
\begin{aligned}
x(t) &= i(t)R + L\frac{di}{dt} \\
x(t) &= i(t)56 + 10 \times 10^{-3} Di(t) \\
i(t) &= \frac{1}{L}\int_{-\infty}^{t} y(\tau)\ d\tau \\
\therefore i(t) &= \frac{1}{L} \times \frac{1}{D} \times y(t)
\end{aligned}
$$

$$
\begin{aligned}
\therefore x(t) &= 56 \times \frac{1}{10 \times 10^{-3}} \times \frac{1}{D} y(t) + \frac{10 \times 10^{-3} D}{10 \times 10^{-3} D} y(t) \\
x(t) &= 56 \times \frac{100}{D} y(t) + y(t) \\
Dx(t) &= 5600 y(t) + Dy(t)
\end{aligned}
$$

$$m = 1\ N = 1 \text{ so } b_0 \neq 0$$

$$\begin{aligned}
\therefore P(D) &= D \\
D + 5600 &= 0 \\
D &= -5600 \\
\lambda &= -5600
\end{aligned}$$

$$\begin{aligned}
\therefore y_n(t) &= e^{-5600t} \\
y_n^{(1-1)}(0) = 1 &= C_1 e^{-5600(0)} \\
1 &= C_1 e^0 \\
\therefore C_1 &= 1
\end{aligned}$$

$$\begin{aligned}
h(t) &= b_0\delta(t) + (P(D)y_n(t))\, u(t) \ (b_0 = 1) \\
\therefore h(t) &= \delta(t) - 5600e^{-5600t} \times u(t) \\
\therefore H(\omega) &= \int_{-\infty}^{\infty} h(t)e^{-j\omega}\ dt
\end{aligned}$$

Using Laplace transform:

$$\begin{aligned}
H(\omega) &= 1 - \frac{5600}{j\omega + 5600} \\
H(f) &= 1 - \frac{5600}{j(2\pi f) + 5600} \\
|H(f)| &= \sqrt{1 + \frac{(5600)^2}{(2\pi f)^2 + (5600)^2}}
\end{aligned}$$

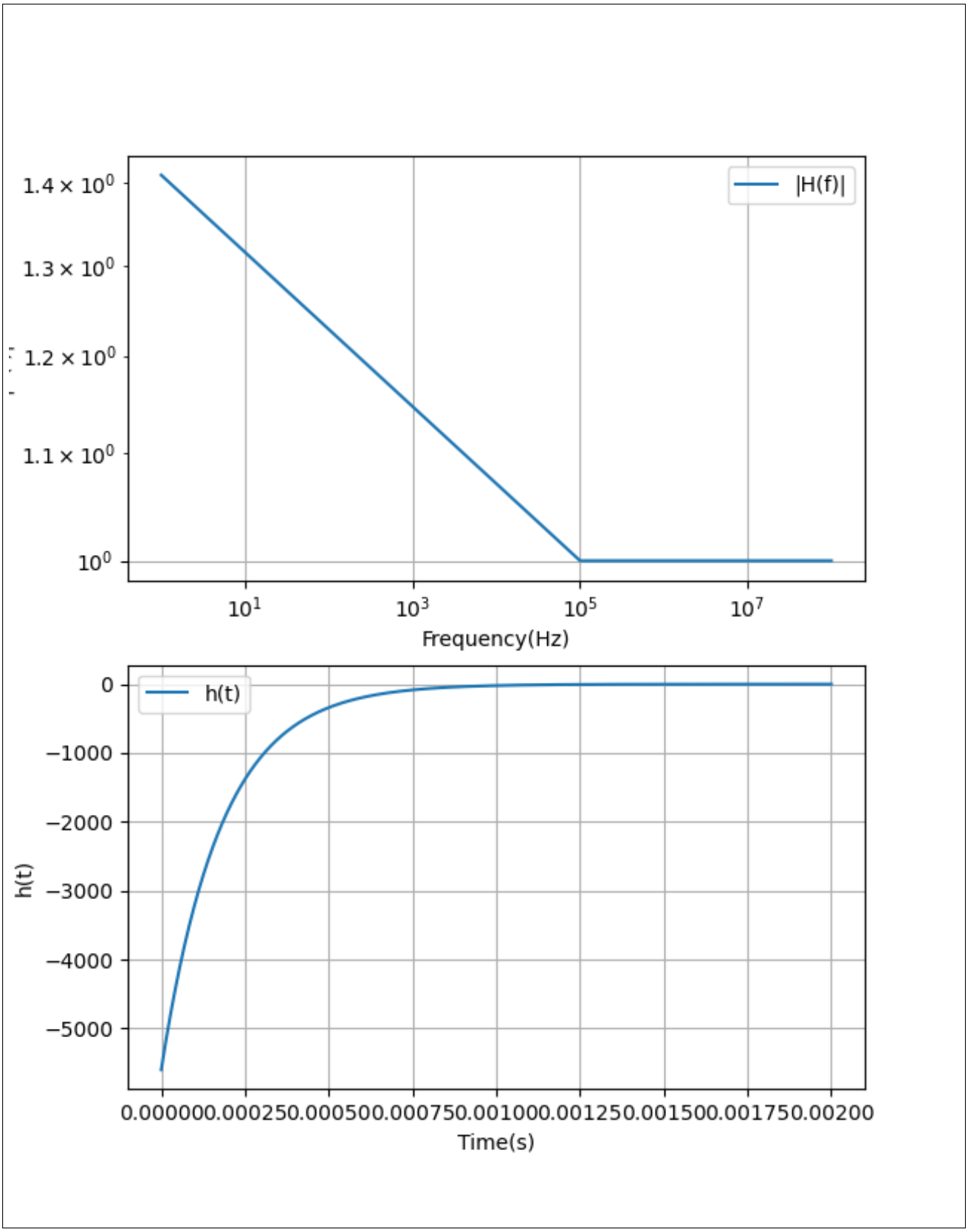

Figure 2.2.4: Plotting of ii

iii)

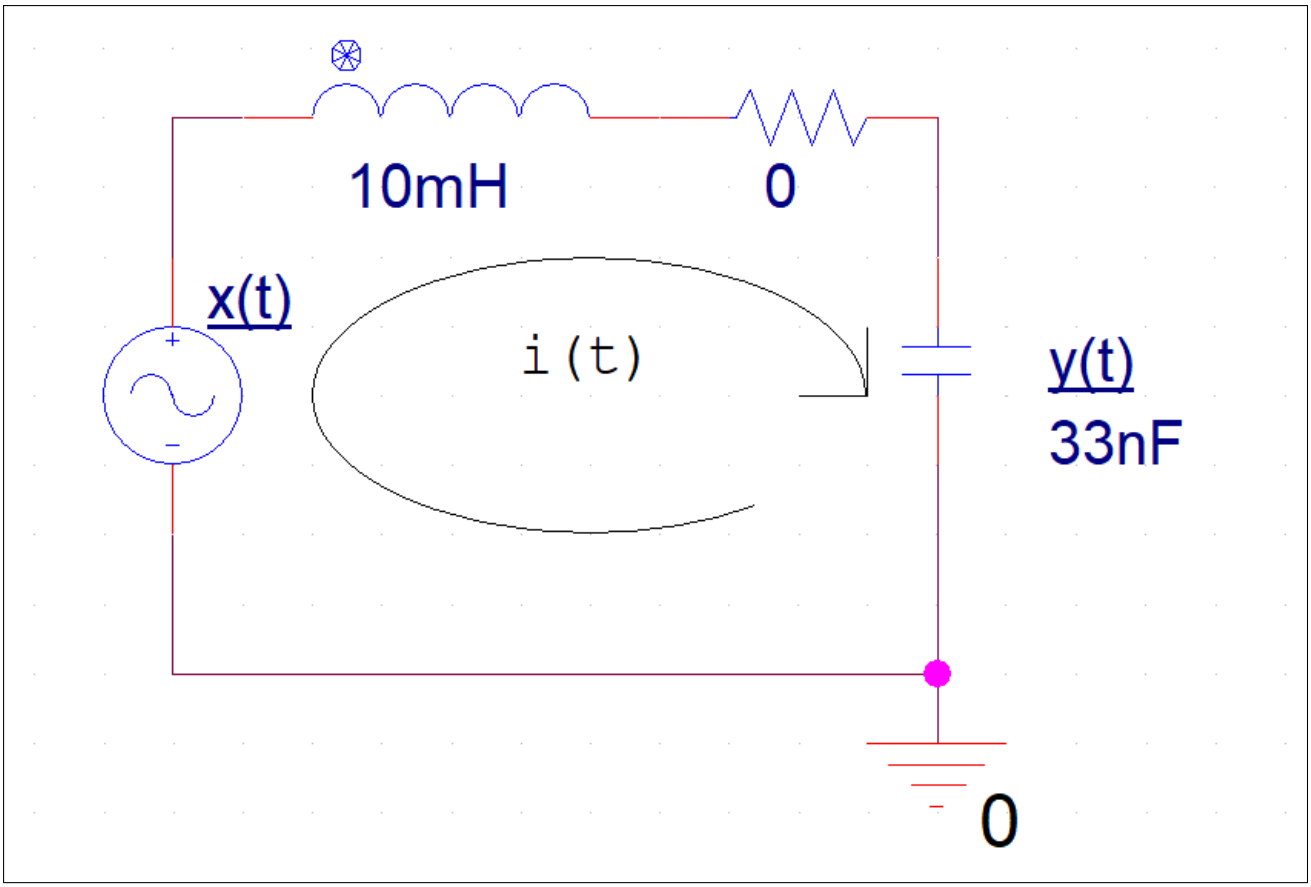

Figure 2.2.5: Circuit used in (iii) calculations

$$
\begin{aligned}
x(t) &= L\frac{di(t)}{dt} + R_{RLC}i(t) + y(t) \\
i(t) &= C\frac{dv}{dt} \\
i(t) &= CDy(t) \\
x(t) &= LDi(t) + R_{RLC}i(t) + y(t) \\
x(t) &= LD(CDy(t)) + R_{RLC}(CDy(t)) + y(t) \\
x(t) &= D^2LCy(t) + DR_{RLC}Cy(t) + y(t) \\
\therefore x(t) &= y(t)\left(D^2LC + DR_{RLC}C + 1\right) \\
P(D) &= 1
\end{aligned}
$$

$$
\begin{aligned}
D^2(10\times10^{-3})(33\times10^{-9}) + D(0)(33\times10^{-9}) + 1 &= 0 \\
D^2 3.3\times10^{-10} + 1 &= 0 \\
D^2 3.3\times10^{-10} &= -1
\end{aligned}
$$

$$
\begin{aligned}
D^2 &= \frac{-1}{3.3\times10^{-10}} \\
D^2 &= -3030303030
\end{aligned}
$$

$$\therefore D = 55048.19j \;\; D = -55048.19j$$

$$\lambda = 55048.19j$$

$$\therefore y_n = C_1 e^{-55048.19t} + C_2 e^{55048.19t}$$

$$C_1 = \tfrac{C}{2} e^{j\theta} \text{ and } C_2 = \tfrac{C}{2} e^{-j\theta}$$

$$\therefore y_n(t) = \frac{C}{2} e^{\alpha t} \left(e^{j(\beta t + \theta)} + e^{-j(\beta t + \theta)} \right)$$

$$\therefore y_n(t) = C e^{\alpha t} cos(\beta t + \theta)$$
$$\therefore y_n(t) = C \times cos(55048.19t + \theta)$$

$N = 2$ second order derivative

$$y^{(1-1)}(0) = 0$$

$$\text{y}^{(2-1)}(0) = 1$$
$$\therefore 0 = C \; cos(55048.19t + \theta)$$
$$\text{y'}_n(t) = 55048.19 \times C sin(55048.19t + \theta)$$

$$\text{When } \theta = tan^{-1}\left(\frac{1}{0}\right)$$
$$\text{thus: } \theta = \lim_{x \to \infty} arctan(x) = \frac{\pi}{2}$$
$$\theta = \frac{\pi}{2}$$

$$y_n'(0) = 1 = -55048.19 C sin\left(0 + \frac{\pi}{2}\right)$$
$$C = -\frac{1}{55048.19}$$
$$C = -1.8165 \times 10^{-5}$$

$$\therefore y_n(t) = -1.8165 \times 10^{-5} cos\left(55048.19t + \frac{\pi}{2}\right)$$

$$H(s) = \frac{\left(-1.8165 \times 10^{-5} cos\frac{\pi}{2}\right) s + ((0)rcos\theta - 55048.19(-1.8165 \times 10^{-5}))}{s^2 + 2(0)s + (a^2 + (55048.19)^2)}$$

$$H(s) = \frac{(-55048.19)(-1.8165 \times 10^{-5})}{s^2 + (55048.19)^2}$$

$$H(s) = \frac{1}{s^2 + (55048.19)^2}$$

$$H(\omega) = \frac{1}{(j\omega)^2 + (55048.19)^2}$$

$$H(f) = \frac{1}{(j2\pi f)^2 + (55048.19)^2}$$

$$|H(f)| = \frac{1}{\sqrt{(j2\pi f)^4 + (55048.19)^4}}$$

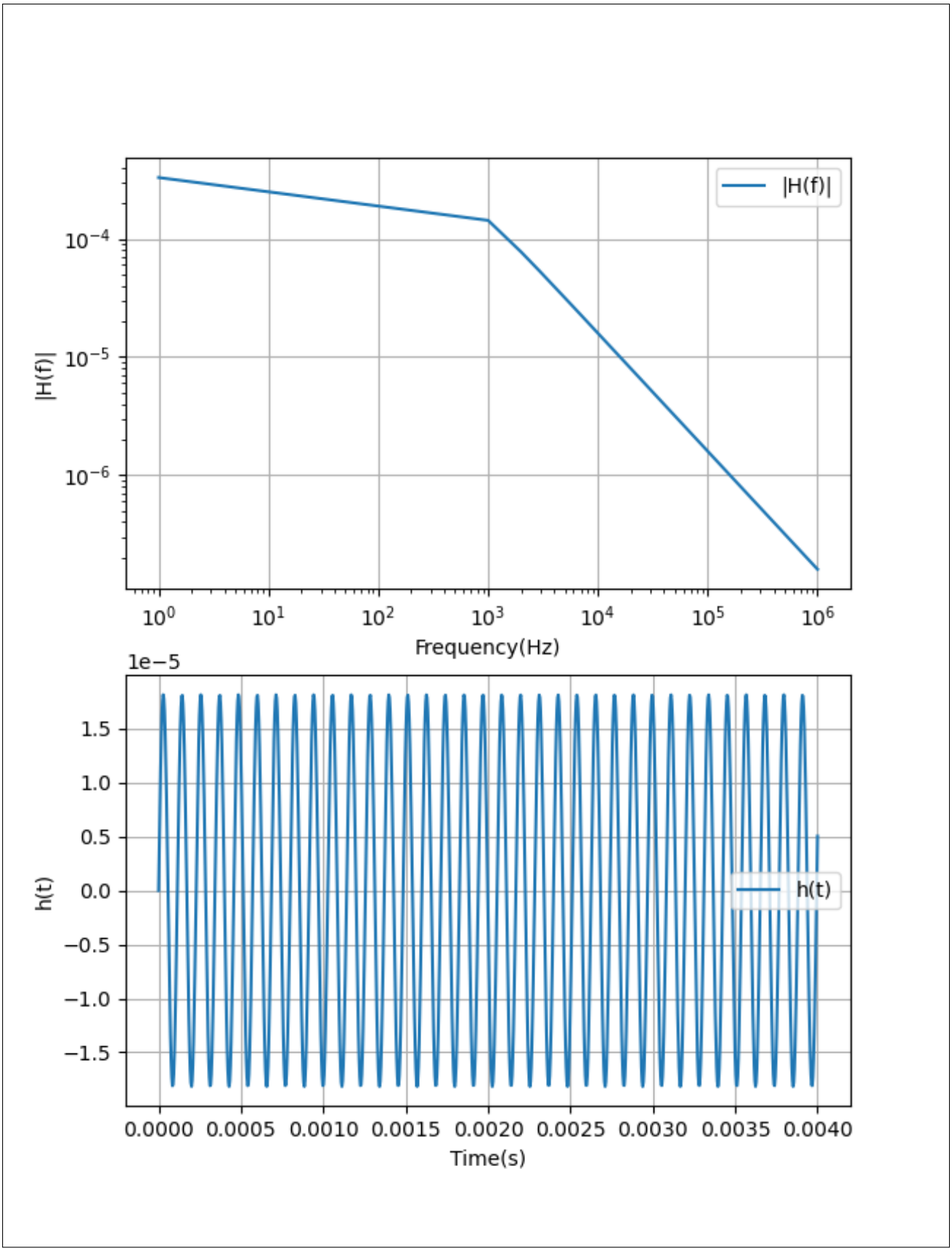

Figure 2.2.6: Plotting of iii

iv)

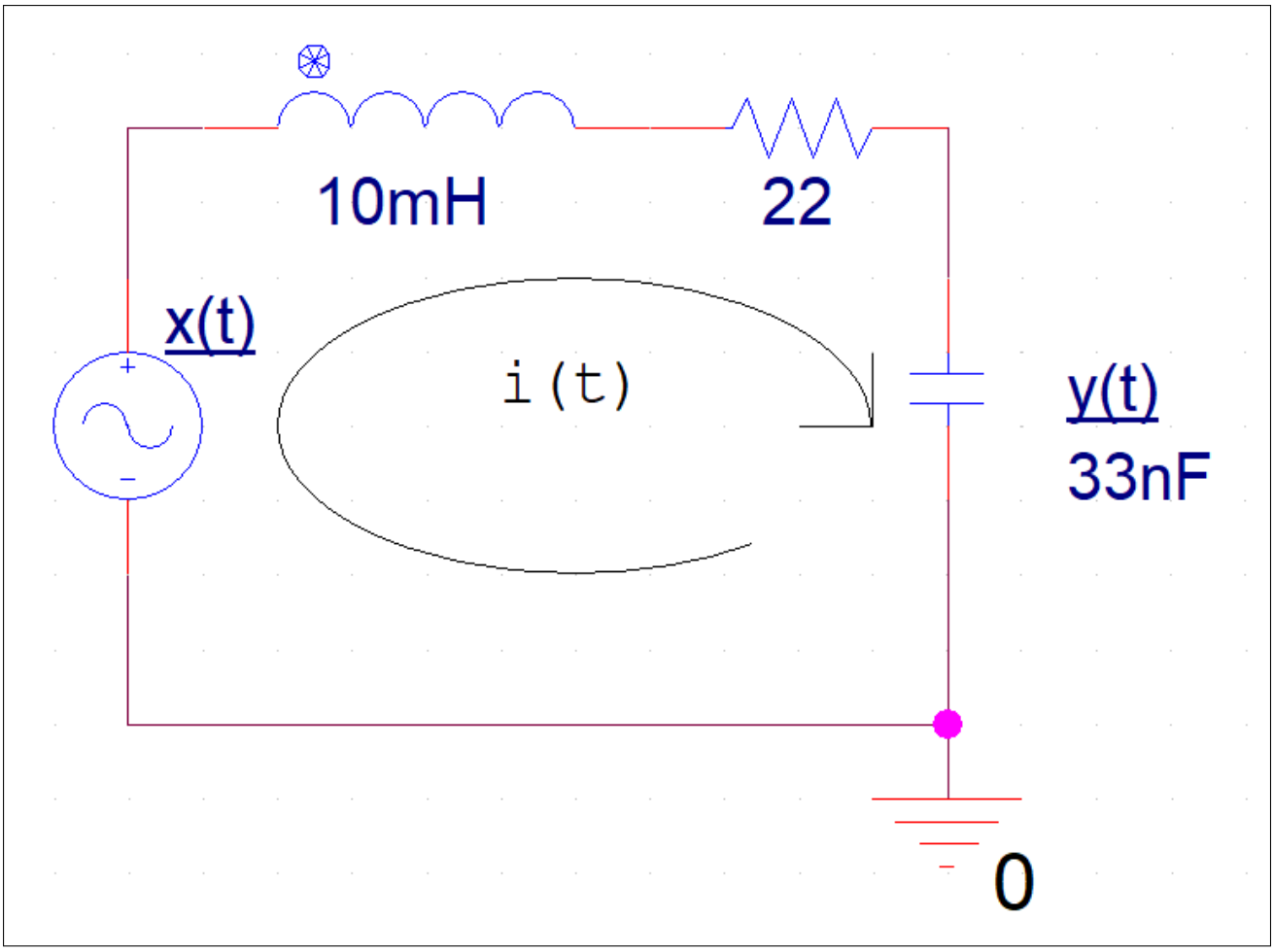

Figure 2.2.7: Circuit used in (iv) calculations

$$
\begin{aligned}
x(t) &= L\frac{di(t)}{dt} + R_{RLC}i(t) + y(t) \\
i(t) &= C\frac{dv}{dt} \\
i(t) &= CDy(t) \\
x(t) &= LDi(t) + R_{RLC}i(t) + y(t) \\
x(t) &= LD(CDy(t)) + R_{RLC}(CDy(t)) + y(t) \\
x(t) &= D^2LCy(t) + DR_{RLC}Cy(t) + y(t) \\
\therefore x(t) &= y(t)\left(D^2LC + DR_{RLC}C + 1\right) \\
P(D) &= 1
\end{aligned}
$$

$$
\begin{aligned}
\therefore D^2(10\times10^{-3})(33\times10^{-9}) + D(22)(33\times10^{-9}) + 1 &= 0 \\
3.3\times10^{-10}D^2 + 7.26\times10^{-7}D + 1 &= 0 \\
3.3\times10^{-10}\lambda^2 + 7.26\times10^{-7}\lambda + 1 &= 0
\end{aligned}
$$

$$
\lambda = \frac{-(7.26\times10^{-7} \pm \sqrt{(7.26\times10^{-7})^2 - 4(3.3\times10^{-10})(1)}}{2(3.3\times10^{-10})}
$$

$$\lambda = -1100 + j55037.1967 \; \lambda = -1100 - j55037.1967$$
$$\therefore y_n(t) = C_1 e^{(-1100+j55037.19)t} + C_2 e^{(-1100-j55037.19)t}$$
$$y_0(t) = \frac{C}{2} e^{j\theta} e^{(\alpha+j\beta)t} + \frac{C}{2} e^{-j\theta} e^{(\alpha-j\beta)t}$$
$$= \frac{C}{2} e^{\alpha t} \left(e^{j(\beta t+\theta)} + e^{-j(\beta t+\theta)} \right)$$
$$y_n(t) = Ce^{-1100t} cos(55037.19t + \theta)$$

$$y_n(0) = 0 \; y_n'(0) = 1$$
$$y_n(t) = Ce^{-1100t} cos(55037.19t + \theta)$$
$$0 = Ccos(\theta)$$

$$y_n'(t) = Ce^{-1100t} cos(55037.19t + \theta) - Ce^{-1100t} 55037.19 sin(55037.19)$$

$$\theta = tan^{-1}\left(\frac{1}{0}\right) \therefore \lim_{x\to\infty} tan^{-1}(x) \qquad = \frac{\pi}{2}$$
$$\therefore \theta = \frac{\pi}{2}$$

$$1 = -1100Ccos\left(\frac{\pi}{2}\right) - 55037.19Csin\left(\frac{\pi}{2}\right)$$
$$1 = -55037.19CC = -1.8165 \times 10^{-5}$$

$$\therefore y_n(t) = -1.8165 \times 10^{-5} e^{-1100t} cos\left(55037.19t + \frac{\pi}{2}\right)$$
$$\therefore h(t) = -1.8165 \times 10^{-5} e^{-1100t} cos\left(55037.19t + \frac{\pi}{5}\right) u(t)$$

According to Laplace transform table:

$$H(s) = \frac{(-1.8165 \times 10^{-5} cos\left(\frac{\pi}{2}\right) s + \left((-1100)(-1.8165 \times 10^{-5} cos\left(\frac{\pi}{2}\right)\right)}{s^2 - 2200s + (1100)^2 + (55037.19)^2}$$
$$- \frac{(55037.19)(-1.8165 \times 10^{-5}) sin\left(\frac{\pi}{2}\right)}{s^2 - 2200s + (1100)^2 + (55037.19)^2}$$
$$H(s) = \frac{1}{s^2 - 2200s + (1100)^2 + (55037.19)^2}$$
$$H(\omega) = \frac{1}{(j\omega)^2 - 2200(j\omega) + (1100)^2 + (55037.19)^2}$$
$$H(f) = \frac{1}{(j2\pi f)^2 - 2200(j2\pi f) + (1100)^2 + (55037.19)^2}$$
$$|H(f)| = \frac{1}{\sqrt{(2\pi f)^4 - (2200(2\pi f))^2 + (1100)^4 + (55037.19)^4}}$$
$$\therefore |H(f)| = \frac{1}{\sqrt{(2\pi f)^4 + (4400\pi f)^2 + (55048.19)^4}}$$

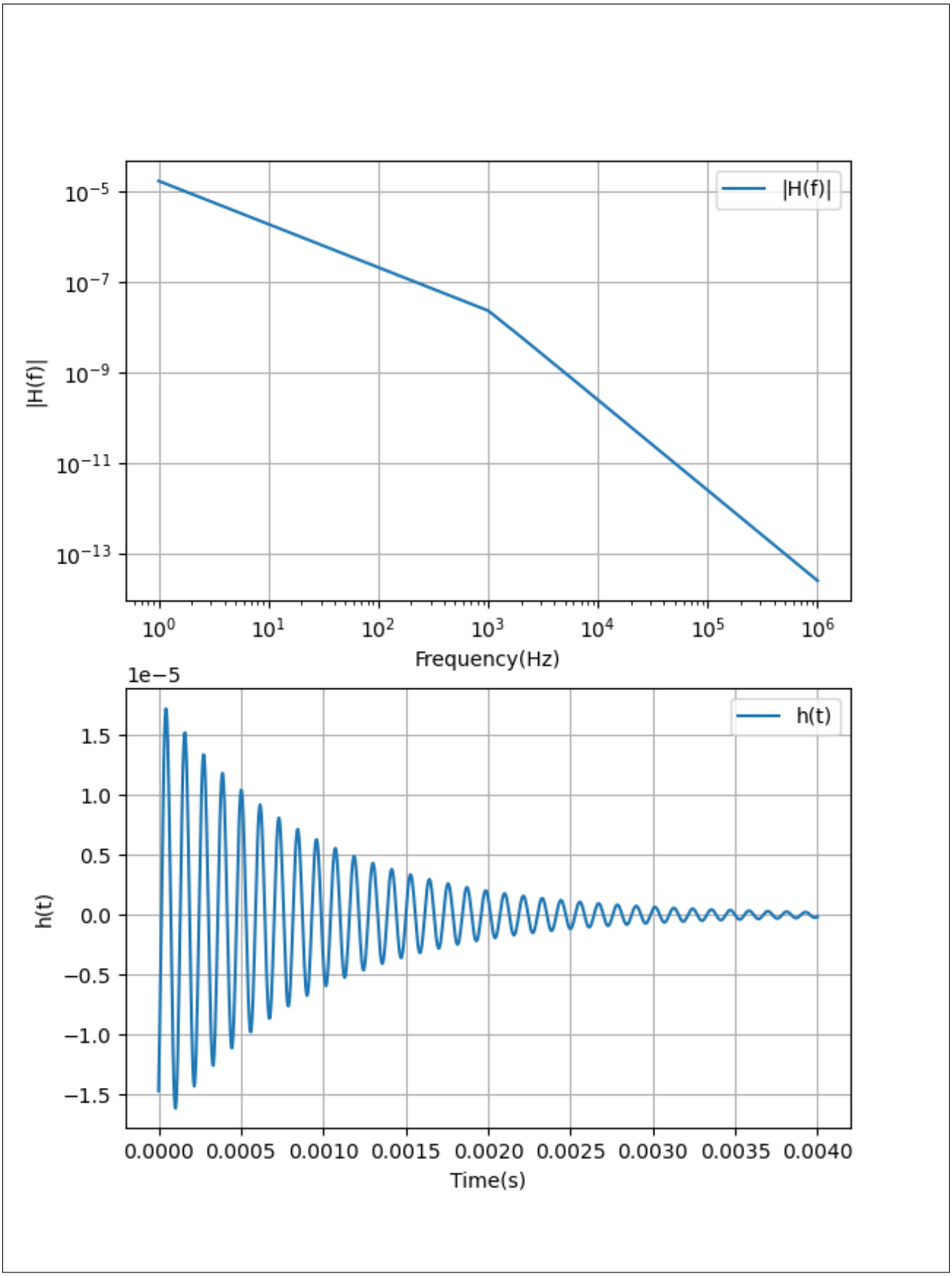

Figure 2.2.8: Plotting of iv

v)

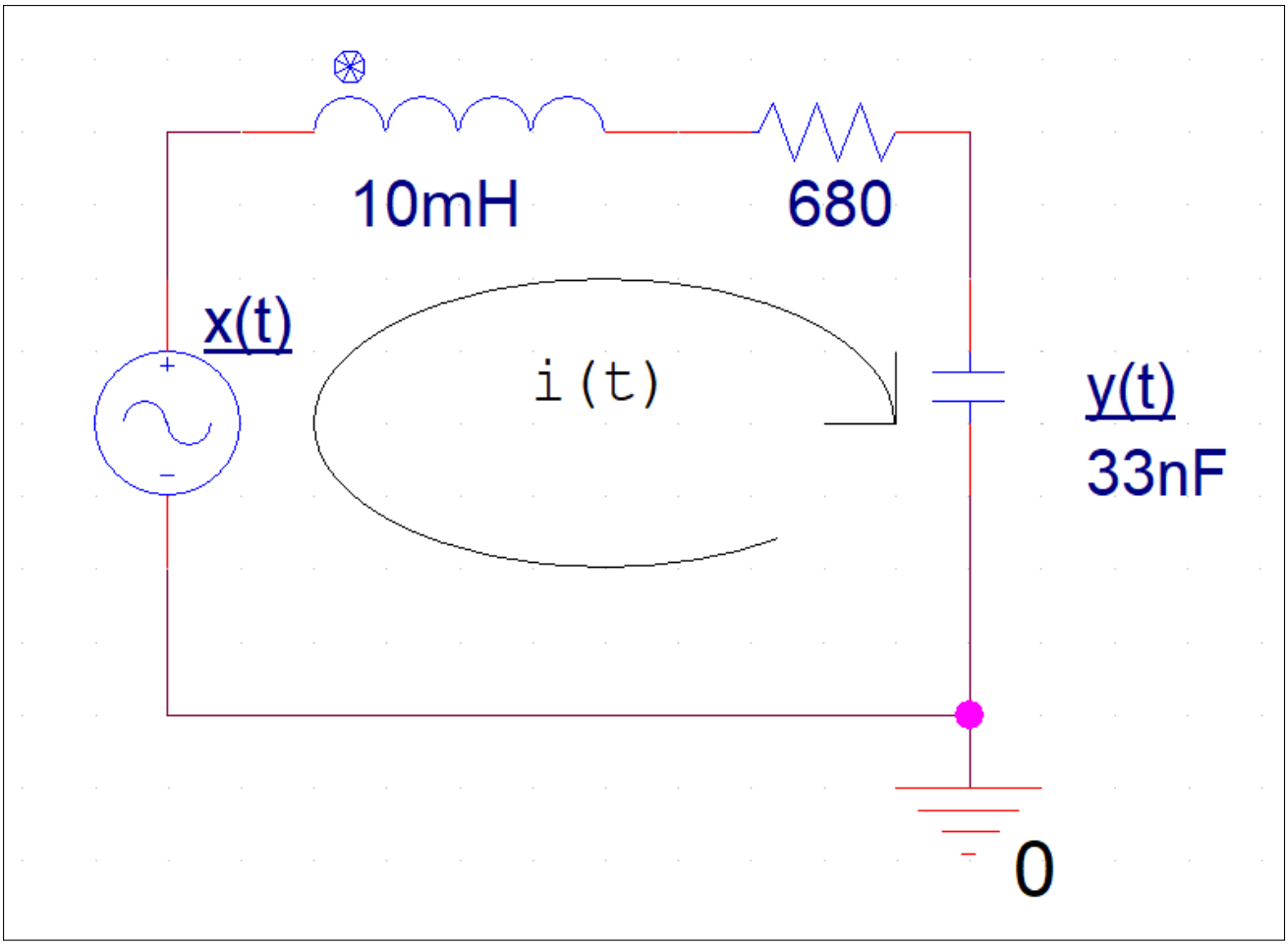

Figure 2.2.9: Circuit used in (v) calculations

$$
\begin{aligned}
x(t) &= L\frac{di(t)}{dt} + R_{RLC}i(t) + y(t) \\
i(t) &= C\frac{dv}{dt} \\
i(t) &= CDy(t) \\
x(t) &= LDi(t) + R_{RLC}i(t) + y(t) \\
x(t) &= LD(CDy(t)) + R_{RLC}(CDy(t)) + y(t) \\
x(t) &= D^2LCy(t) + DR_{RLC}Cy(t) + y(t) \\
\therefore x(t) &= y(t)\left(D^2LC + DR_{RLC}C + 1\right) \\
P(D) &= 1
\end{aligned}
$$

$$
\begin{aligned}
\therefore D^2(10\times10^{-3})(33\times10^{-9}) + D(680)(33\times10^{-9}) + 1 &= 0 \\
3.3\times10^{-10}D^2 + 2.244\times10^{-5}D + 1 &= 0
\end{aligned}
$$

$$
\lambda = \frac{-(2.244\times10^{-5} \pm \sqrt{(2.244\times10^{-5})^2 - 4(3.3\times10^{-10})(1)}}{2(3.3\times10^{-10})}
$$

$$\lambda = -34000 + j43293.22 \; \lambda = -34000 - j43293.22$$
$$\therefore y_n(t) = C_1 e^{(-34000-j43293.22)t} + C_2 e^{(-34000+j43293.22)t}$$
$$y_n(t) = Cecos(\beta t + \theta)$$
$$\therefore y_n(t) = Ce^{-34000t} cos(43293.22t + \theta)$$

$$N = 2 \text{ Second order}$$

$$y_n(0) = 1 \; y_n'(0) = 0$$

$$y_n(0) = 0 = Ccos(\theta)$$
$$y_n'(0) = 1$$

$$y_n'(t) = -34000Ce^{-34000t} cos(43293.22t + \theta) - 43293.22Ce^{-34000t} sin(43293.22 + \theta)$$

$$\theta = tan^{-1}\left(\frac{1}{0}\right)$$
$$\therefore \lim_{x\to\infty} tan^{-1}(x) = \frac{\pi}{2}$$
$$\therefore \theta = \frac{\pi}{2}$$

$$1 = -34000Ccos\left(\frac{\pi}{2}\right) - 43293.22Csin\left(\frac{\pi}{2}\right)$$
$$1 = -43293.22CC = -2.309 \times 10^{-5}$$

$$\therefore y_n(t) = -2.309 \times 10^{-5} e^{-34000t} cos\left(43293.22t + \frac{\pi}{2}\right)$$
$$\therefore h(t) = -2.309 \times 10^{-5} e^{-34000t} cos\left(43293.22t + \frac{\pi}{5}\right) u(t)$$

According to Laplace transform table:

$$H(s) = \frac{(-2.309 \times 10^{-5} cos\left(\frac{\pi}{2}\right) s + (-34000cos\left(\frac{\pi}{2}\right) - (43293.22)(-2.309 \times 10^{-5}) sin\left(\frac{\pi}{2}\right)}{s^2 - 2200s + (1100)^2 + (55037.19)^2}$$
$$H(s) = \frac{-(43293.22)(-2.309 \times 10^{-5}}{s^2 + 68000s + (34000)^2 + (43293.22)^2}$$
$$H(\omega) = \frac{1}{(j\omega)^2 + 68000(j\omega) + (34000)^2 + (43293.22)^2}$$
$$H(f) = \frac{1}{(j2\pi f)^2 + 68000(j2\pi f) + (34000)^2 + (43293.22)^2}$$
$$|H(f)| = \frac{1}{\sqrt{(2\pi f)^4 + (68000(2\pi f))^2 + (34000)^4 + (43293.22)^4}}$$

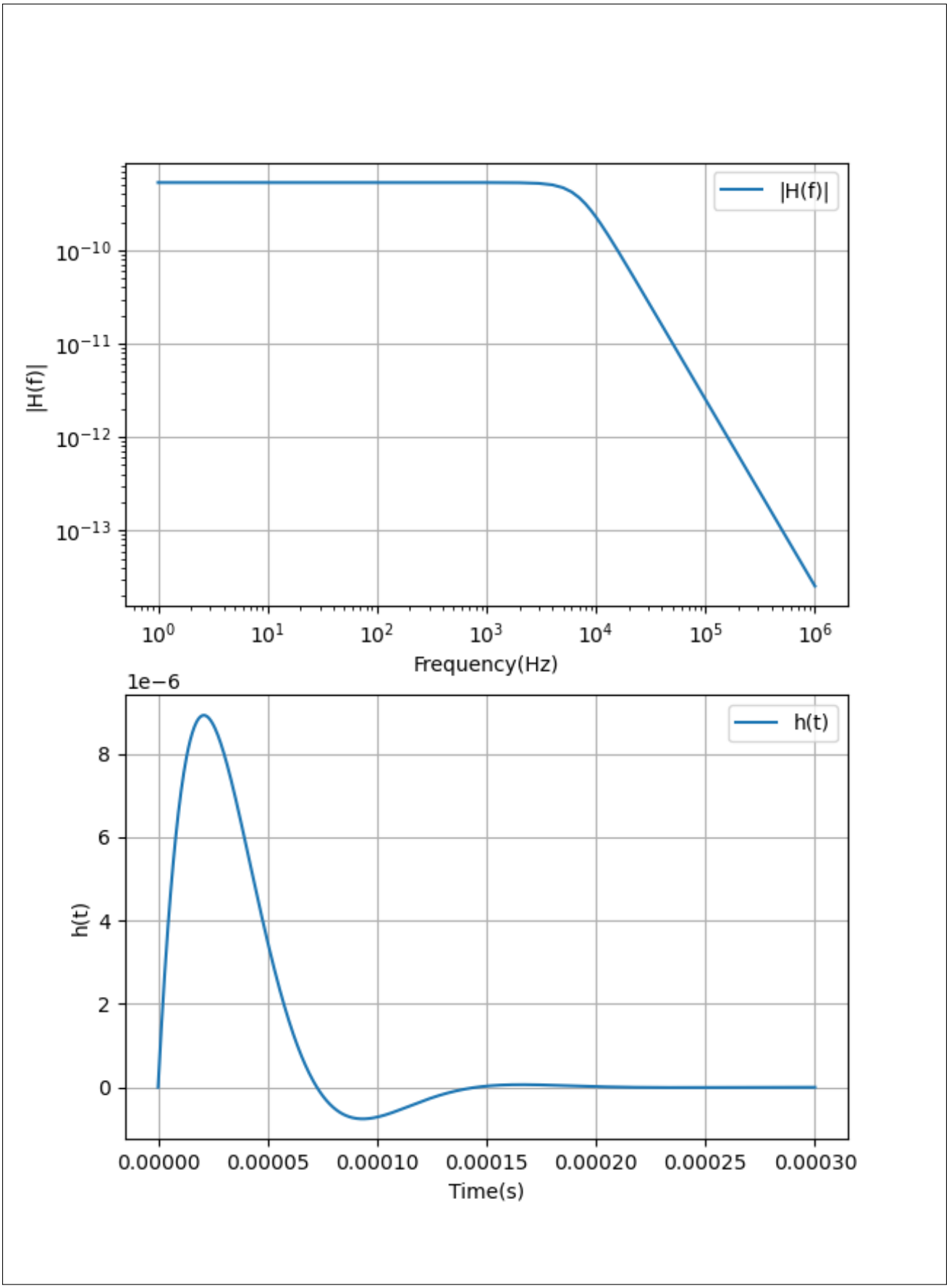

Figure 2.2.10: Plotting of v

vi)

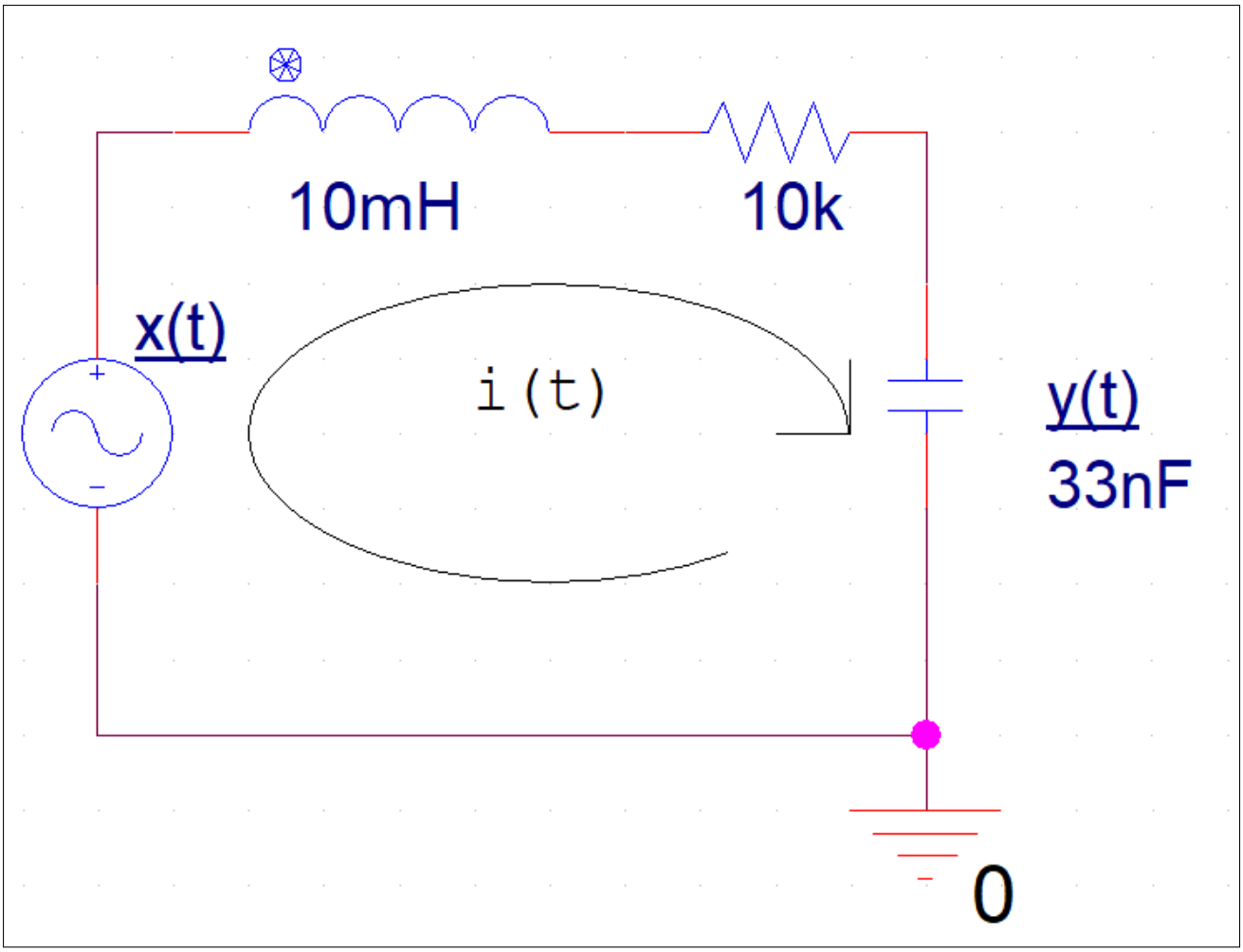

Figure 2.2.11: Circuit used in (vi) calculations

$$
\begin{aligned}
x(t) &= L\frac{di(t)}{dt} + R_{RLC}i(t) + y(t) \\
i(t) &= C\frac{dv}{dt} \\
i(t) &= CDy(t) \\
x(t) &= LDi(t) + R_{RLC}i(t) + y(t) \\
x(t) &= LD(CDy(t)) + R_{RLC}(CDy(t)) + y(t) \\
x(t) &= D^2LCy(t) + DR_{RLC}Cy(t) + y(t) \\
\therefore x(t) &= y(t)\left(D^2LC + DR_{RLC}C + 1\right) \\
P(D) &= 1
\end{aligned}
$$

$$
\begin{aligned}
\therefore D^2(10\times10^{-3})(33\times10^{-9}) + D(10000)(33\times10^{-9}) + 1 &= 0 \\
3.3\times10^{-10}D^2 + 3.3\times10^{-4}D + 1 &= 0
\end{aligned}
$$

$$
\lambda = \frac{-(3.3\times10^{-4} \pm \sqrt{(3.3\times10^{-4})^2 - 4(3.3\times10^{-10})(1)}}{2(3.3\times10^{-10})}
$$

$$
\lambda = -3039.54 \ \lambda = -996960.46
$$

$$
\therefore y_n(t) = C_1e^{-3039.54t} + C_2e^{-996960.46t}
$$

$N = 2$ Second order

$$y_n(0) = 0 \; y_n'(0) = 1$$

$$y_n(0) = C_1 + C_2$$
$$C_1 = -C_2$$

$$\begin{aligned}
y_n'(t) &= -3039.54C_1e^{-3039.54t} - 996960.46C_2e^{-996960.46} \\
y_n'(t) = 1 &= -3039.54C_1 - 996960.46C_2 \\
1 &= -3039.54(-C_2) - 996960.46C_2 \\
1 &= -9939200C_2 \\
C_2 &= -1.006 \times 10^{-6} \\
C_1 &= 1.006 \times 10^{-6}
\end{aligned}$$

$$\begin{aligned}
\therefore y_n(t) &= 1.006 \times 10^{-6}e^{-3039.54t} - 1.006 \times 10^{-6}e^{-996960.46t} \\
\therefore h(t) &= \left(1.006 \times 10^{-6}e^{-3039.54t} - 1.006 \times 10^{-6}e^{-996960.46t}\right) u(t)
\end{aligned}$$

$$\begin{aligned}
H(s) &= 1.006 \times 10^{-6}\left(\frac{1}{s + 3039.54}\right) - 1.006 \times 10^{-6}\left(\frac{1}{s + 996960.46}\right) \\
H(\omega) &= \frac{1.006 \times 10^{-6}}{j\omega + 3039.54} - \frac{1.006 \times 10^{-6}}{j\omega + 996960.46} \\
H(f) &= \frac{1.006 \times 10^{-6}}{j(2\pi f) + 3039.54} - \frac{1.006 \times 10^{-6}}{j(2\pi f) + 996960.46} \\
|H(f)| &= \sqrt{\frac{(1.006 \times 10^{-6})^2}{(2\pi f)^2 + (3039.54)^2} + \frac{(1.006 \times 10^{-6})^2}{(2\pi f)^2 + (996960.46)^2}}
\end{aligned}$$

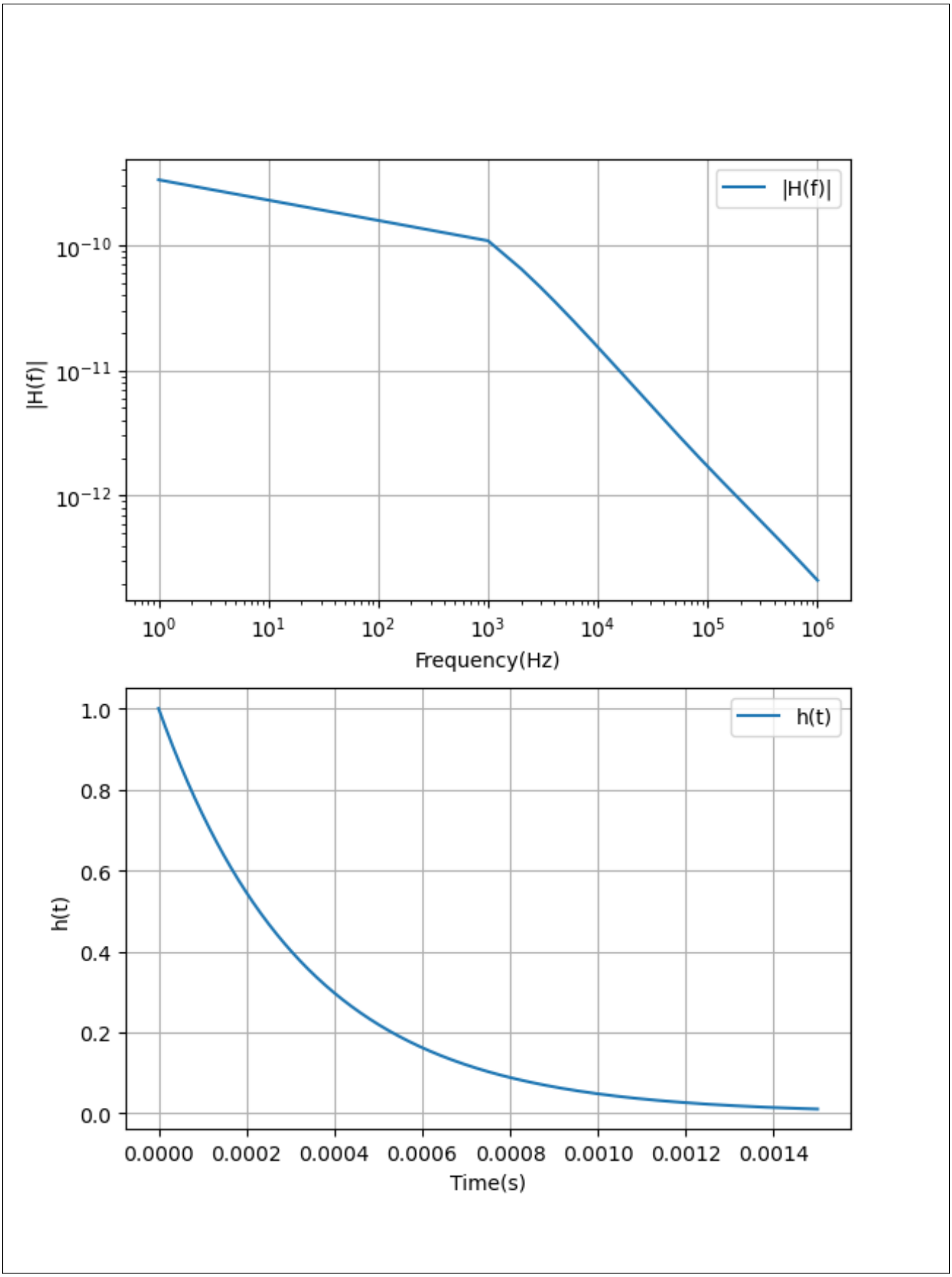

Figure 2.2.12: Plotting of vi

3 Simulation Results

3.1 Question 1

a) Signal generator in time domain with $R_3 = 10k\Omega$

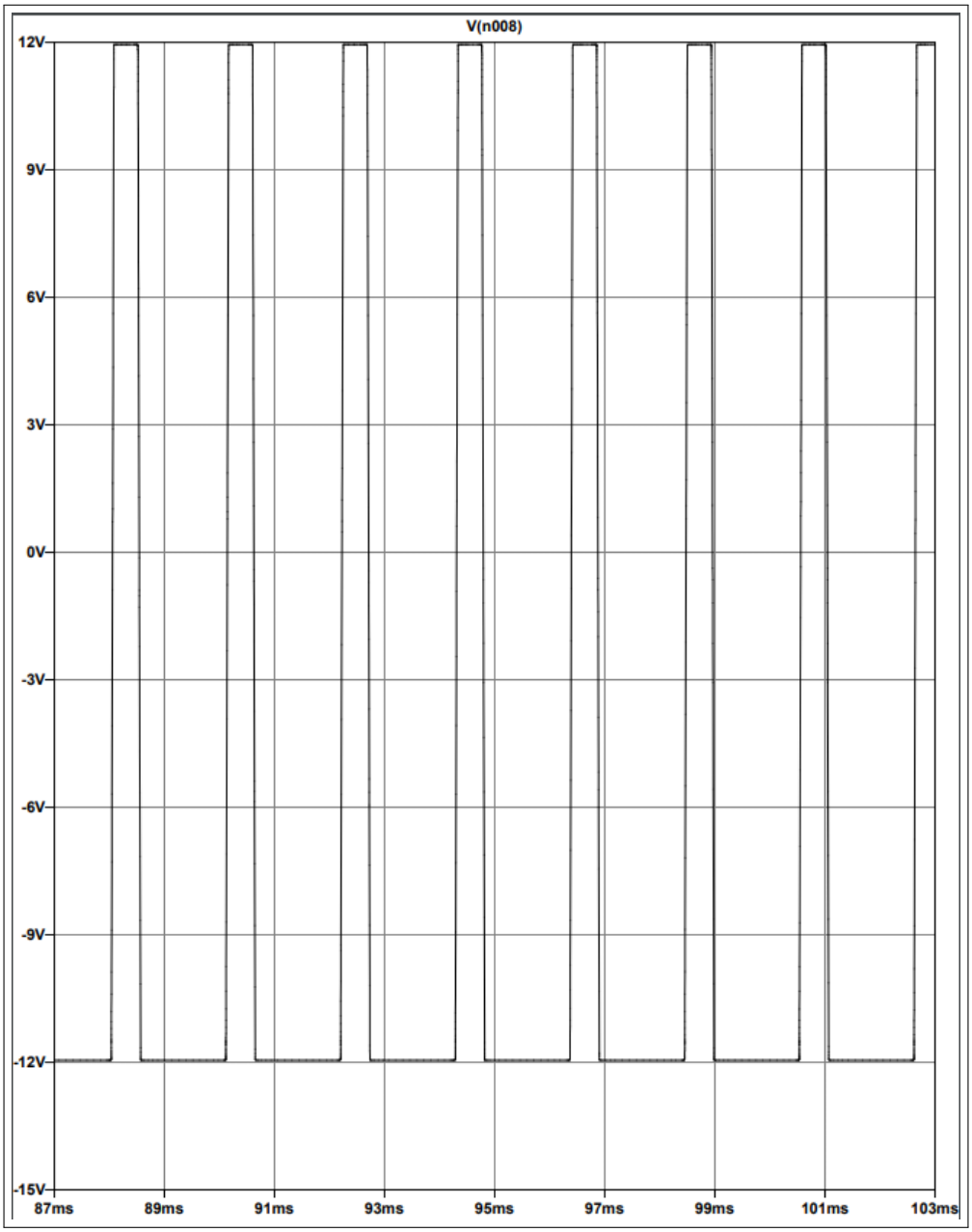

Figure 3.1.1: Signal generator in time domain

b) Signal generator in frequency domain with $R_3 = 10k\Omega$

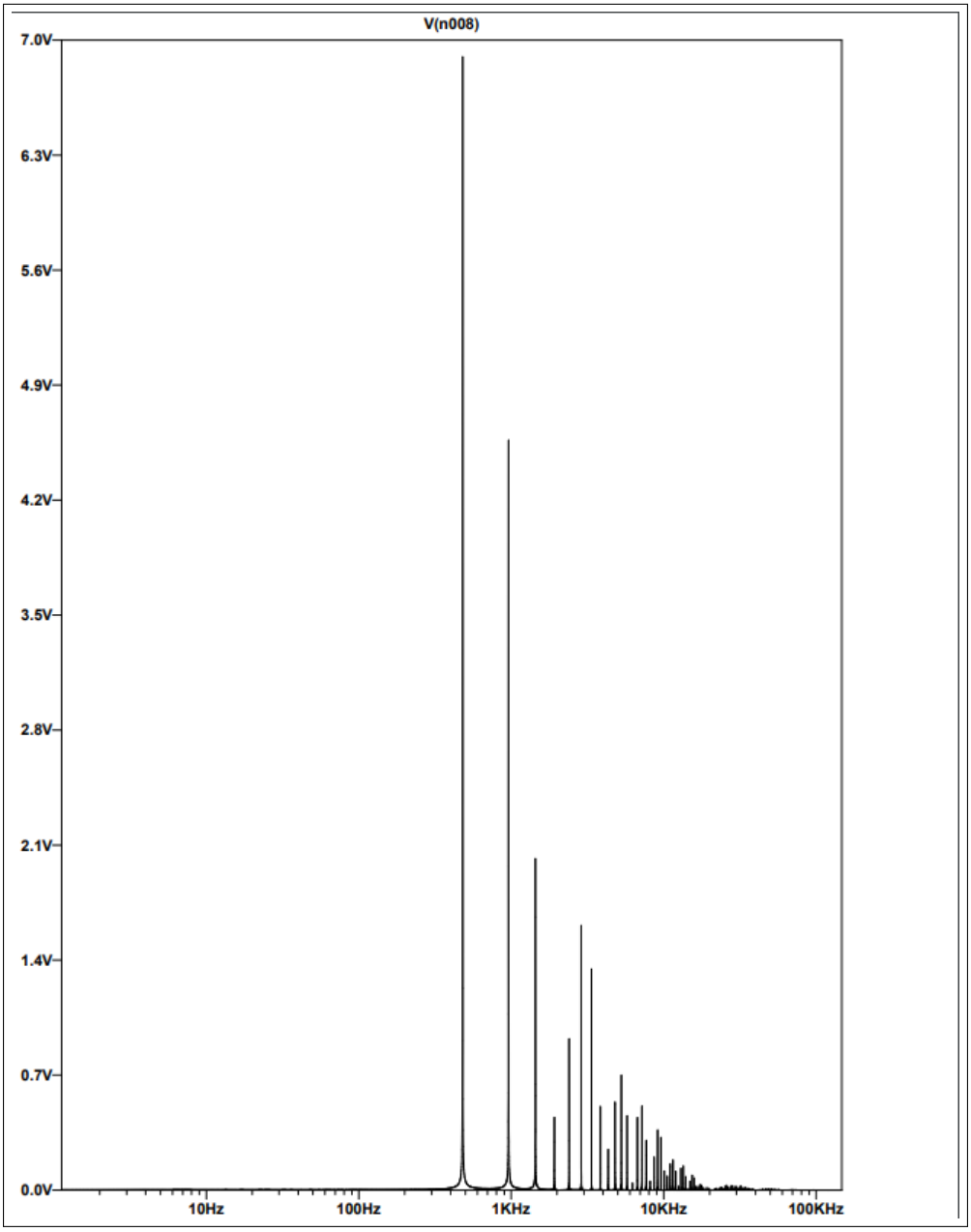

Figure 3.1.2: Signal generator in frequency domain

c) Signal generator in time domain with $R_3 = 1k\Omega$

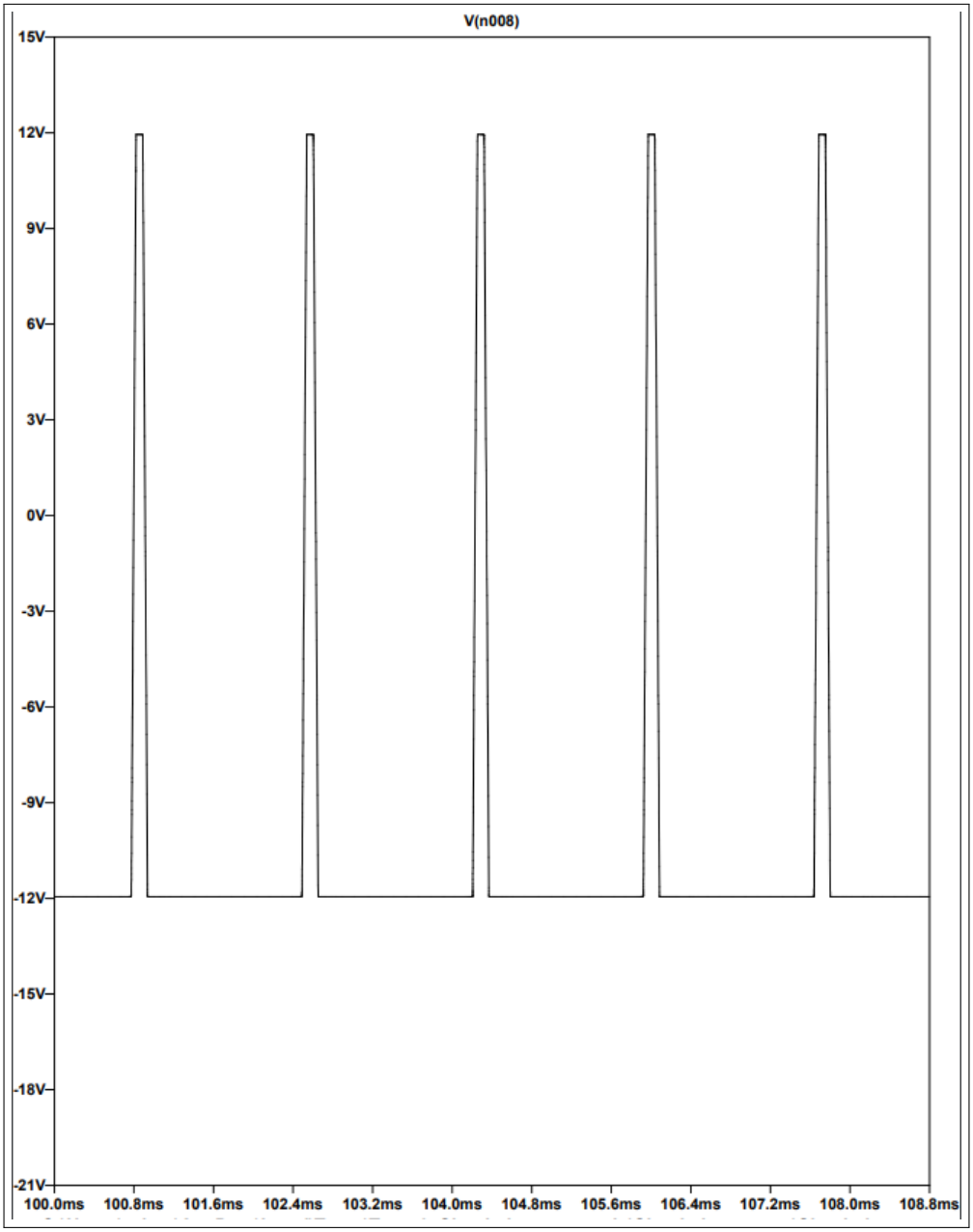

Figure 3.1.3: Signal generator in time domain

Signal generator in frequency domain with $R_3 = 1k\Omega$

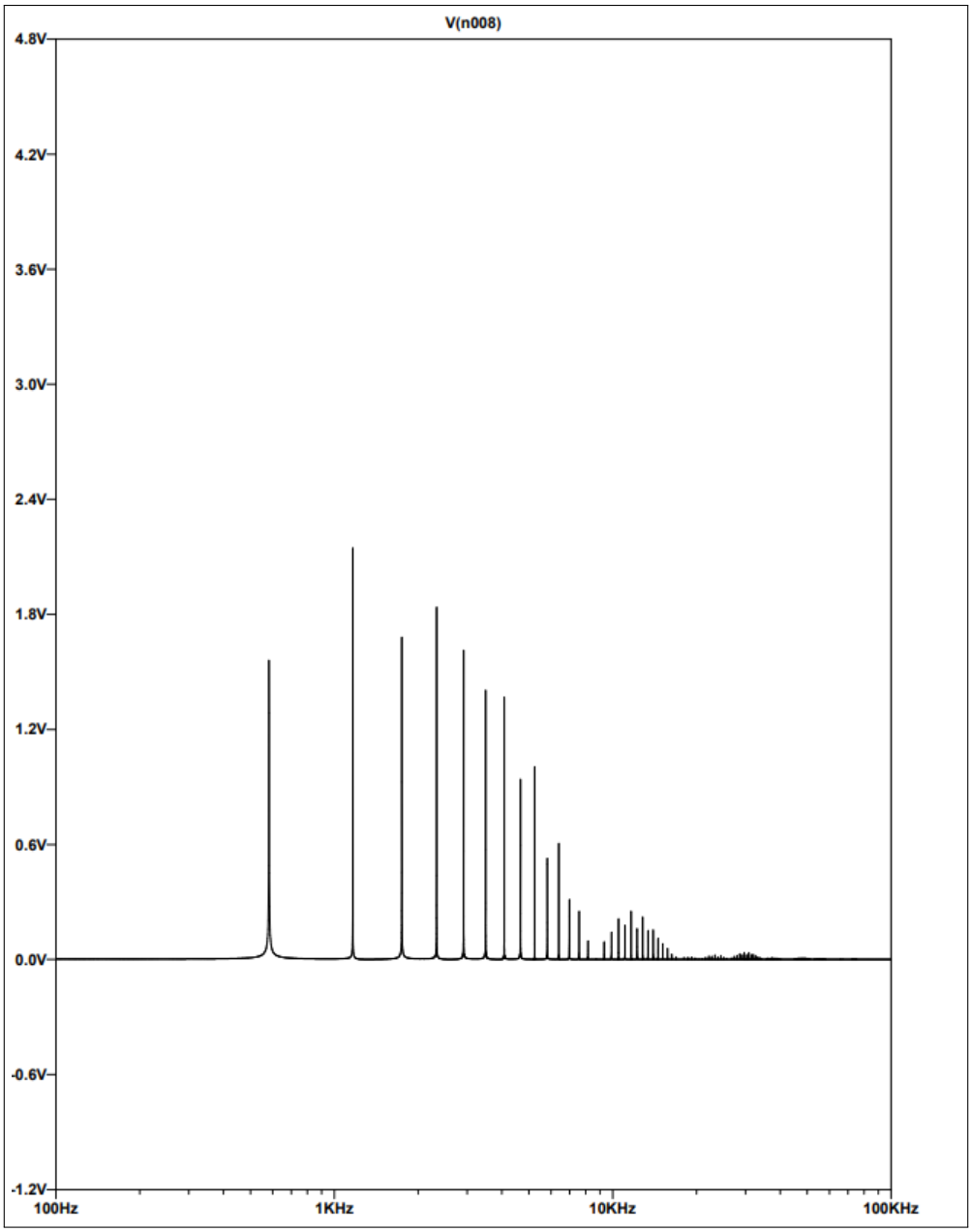

Figure 3.1.4: Signal generator in frequency domain

d) The output of the theoretical circuit that was graphed has a close similarity in the time domain and frequency domain to what was simulated in LTspice. The differences that were noted is the lack of oscillations represented in the frequency domain of the theoretical vs the simulated circuit in LTSpice. The theoretical simulation gives the absolute magnitude of $H(f)$ which accounts for the lack of oscillations in the theoretical circuit vs the oscillations found in the simulated circuit in LTspice.

3.2 Question 2

a) i) $R = 10k\Omega$

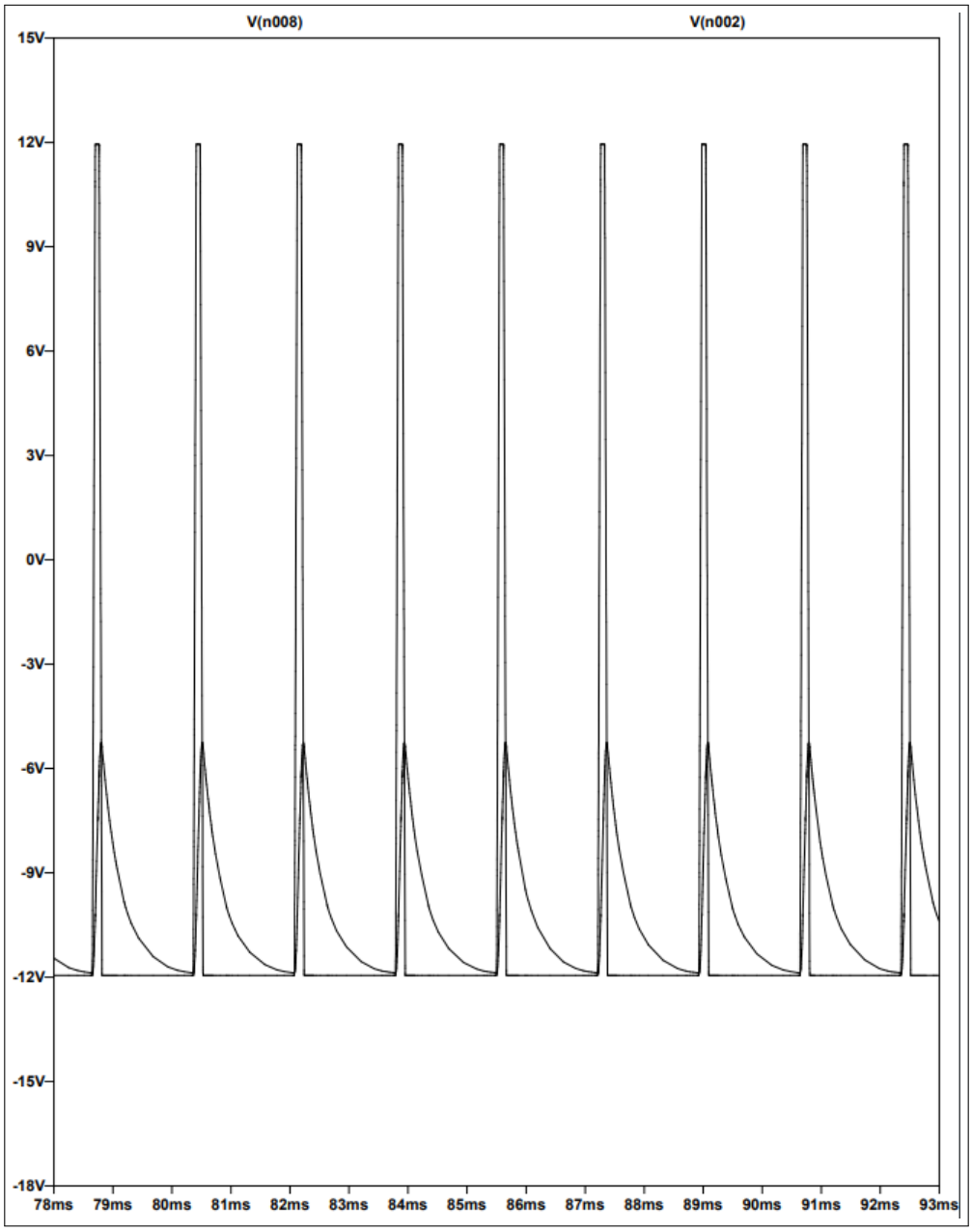

Figure 3.2.1: Signal generator in time domain with only capacitor circuit

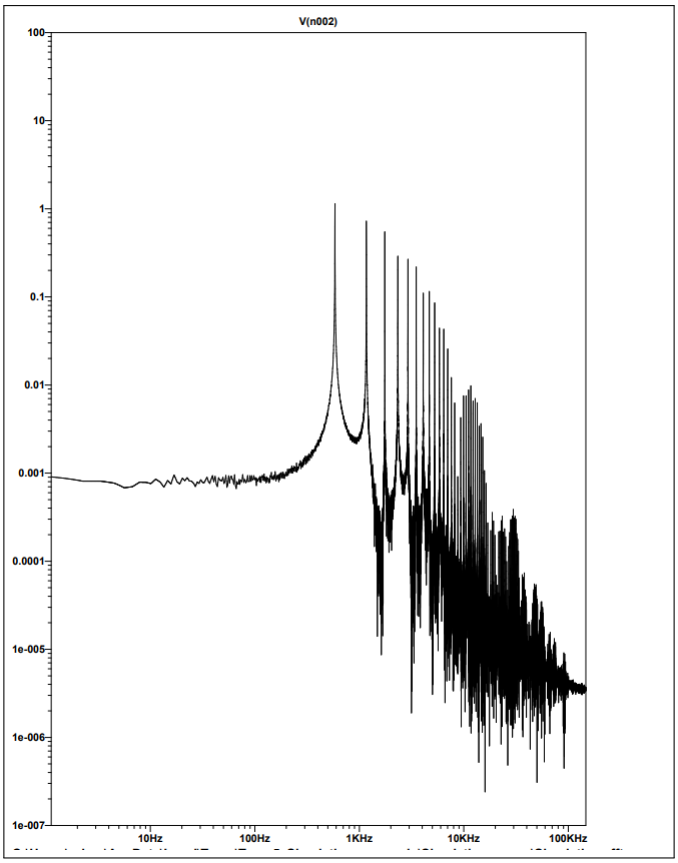

Figure 3.2.2: Logarithmic: Capacitor circuit in frequency domain

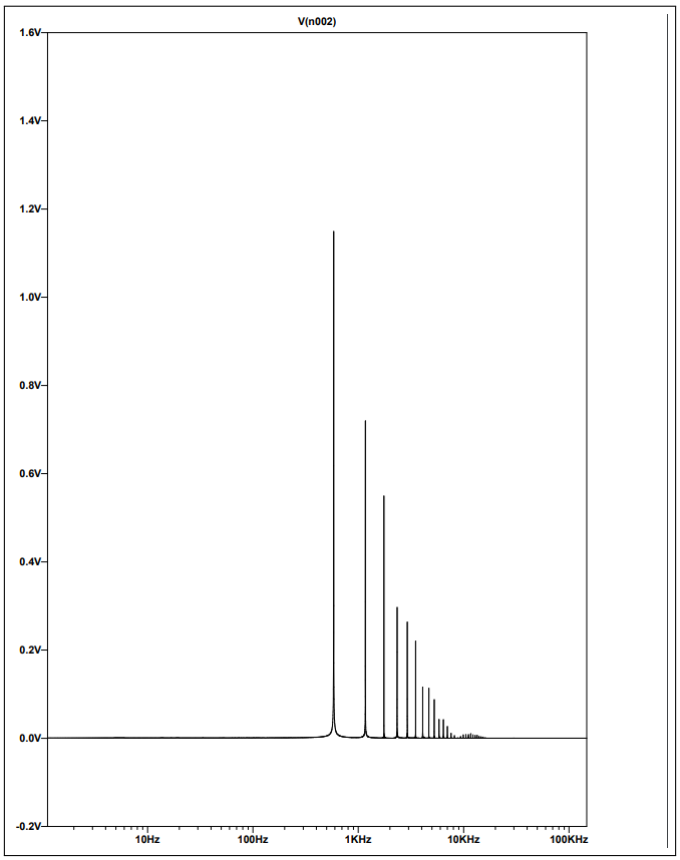

Figure 3.2.3: Linear: Capacitor circuit in frequency domain

ii) $R = 56\Omega$

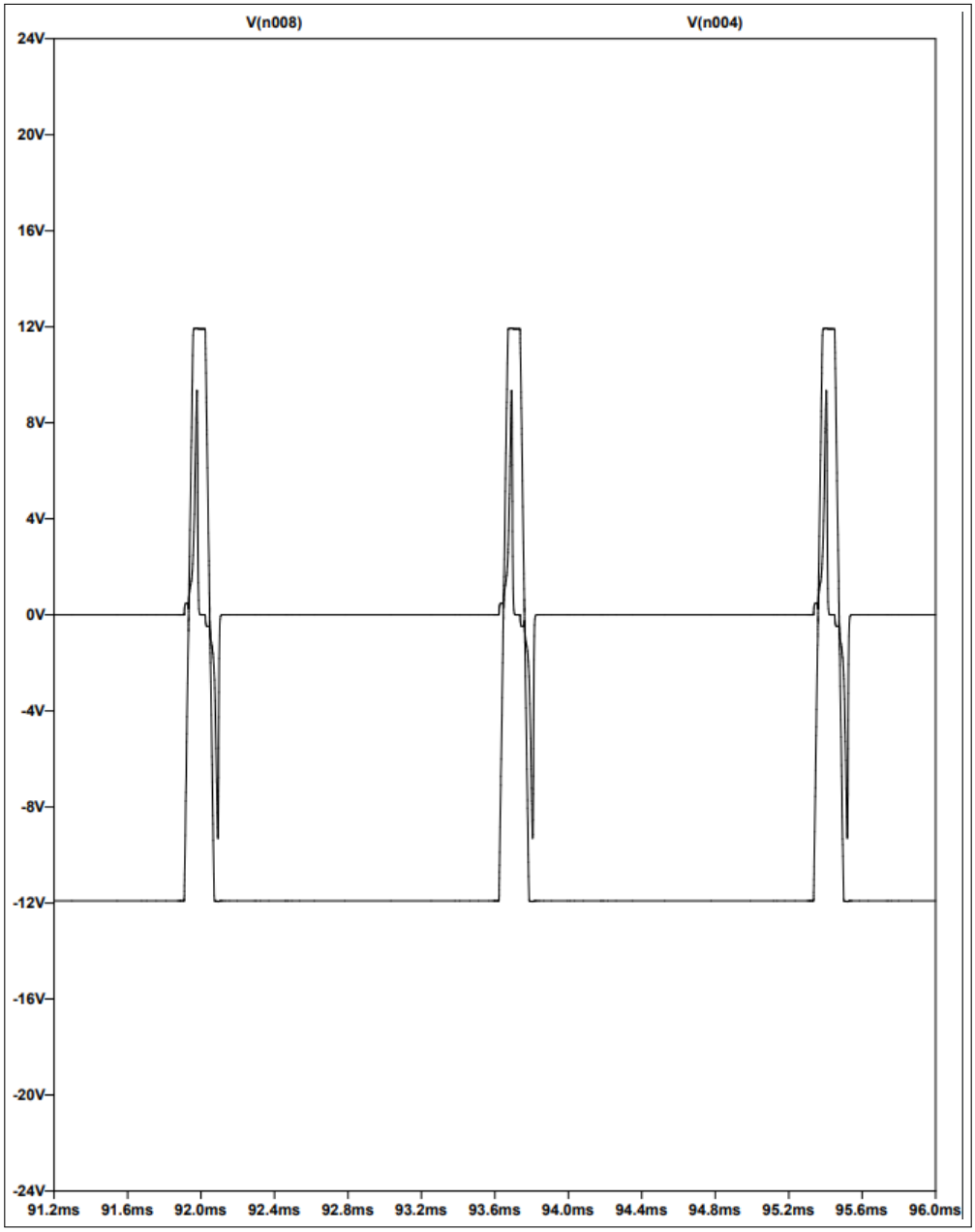

Figure 3.2.4: Signal generator in time domain with only inductor circuit

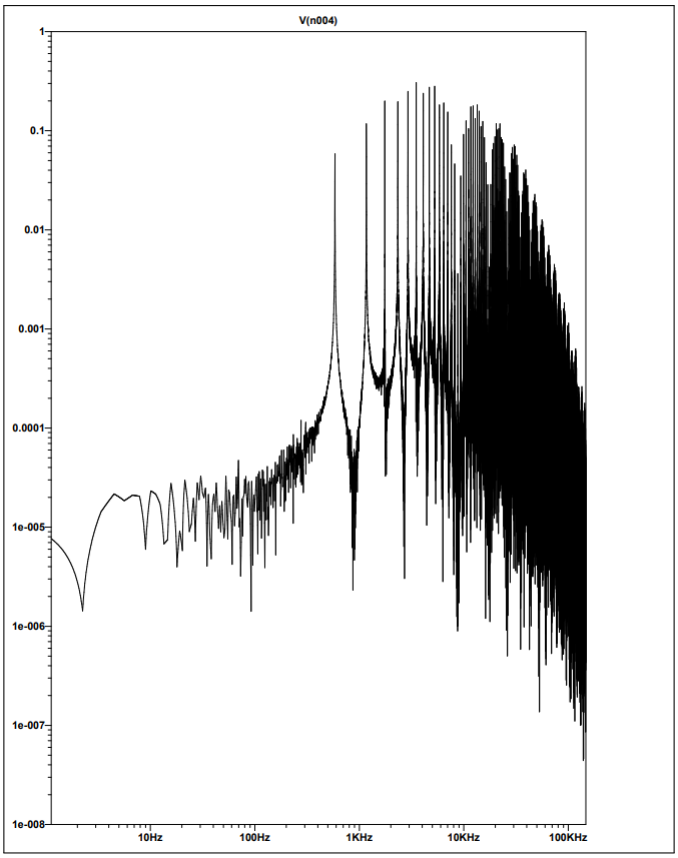

Figure 3.2.5: Logarithmic: Inductor circuit in frequency domain

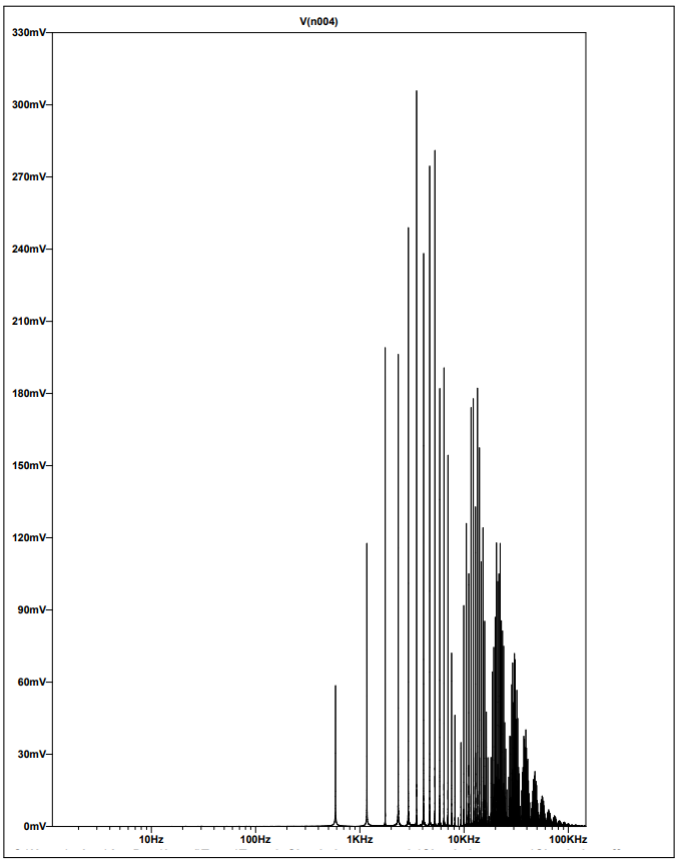

Figure 3.2.6: Linear: Inductor circuit in frequency domain

iii) $R = 0\Omega$

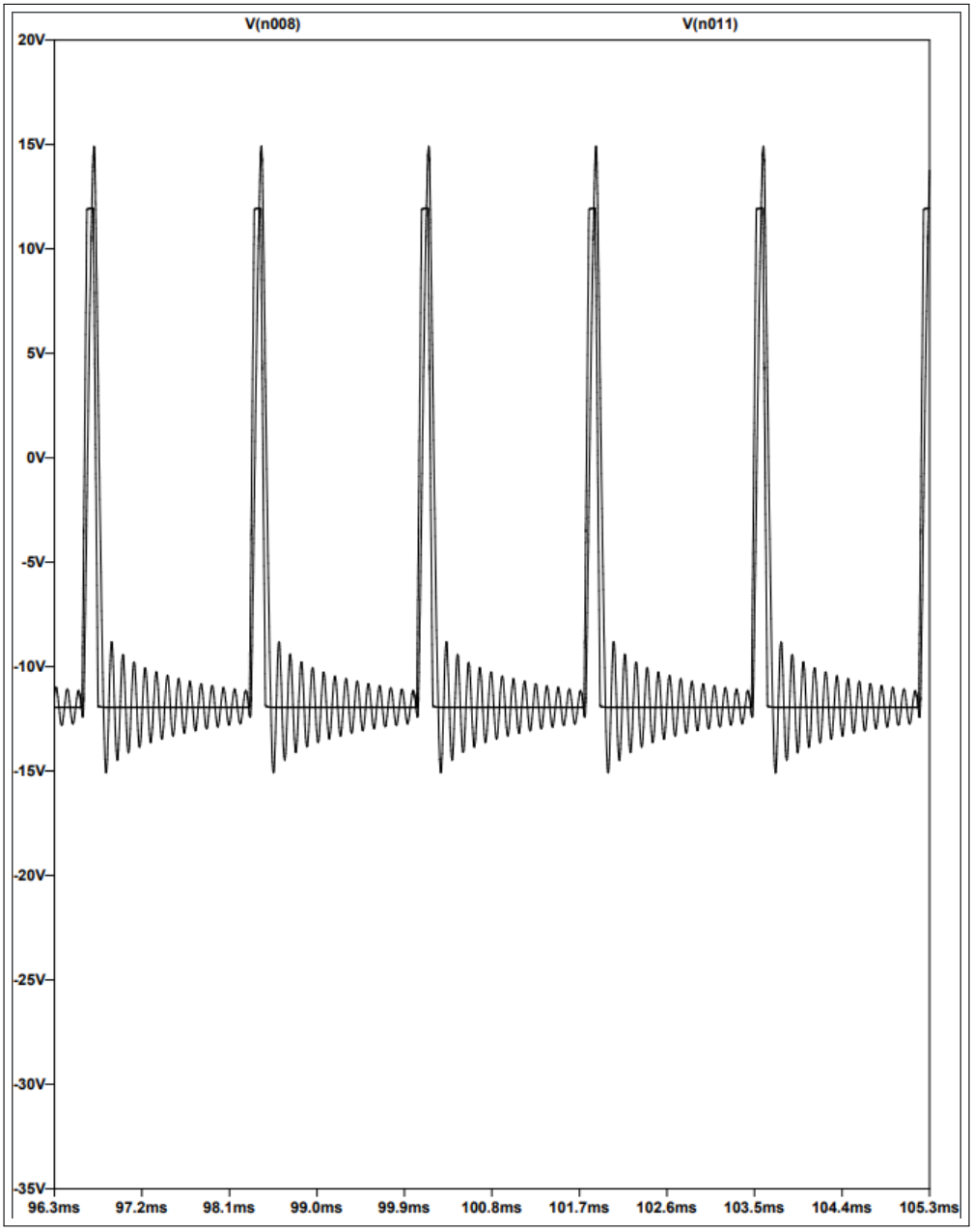

Figure 3.2.7: Signal generator in time domain with inductor and capacitor circuit

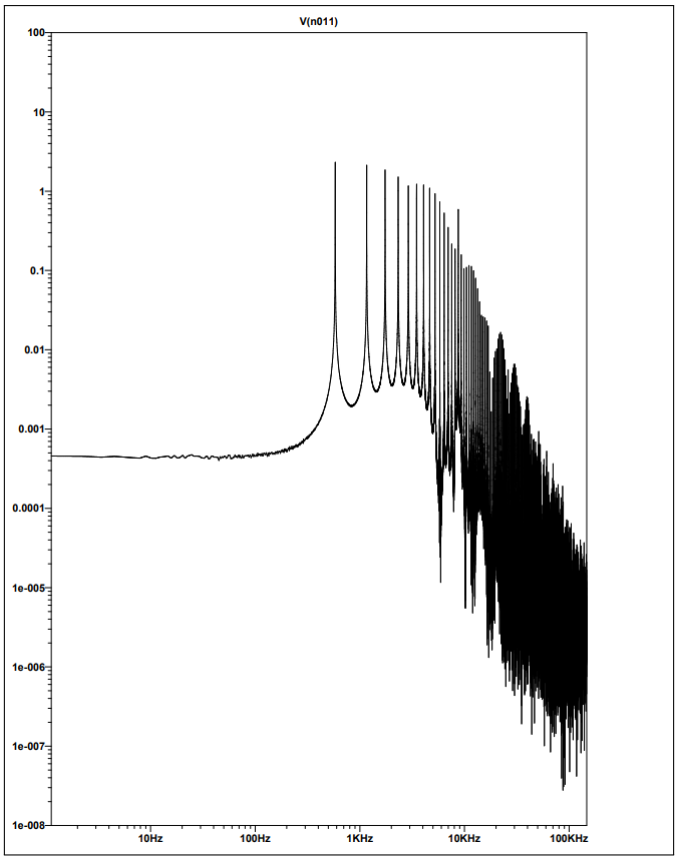

Figure 3.2.8: Logarithmic: Inductor and capacitor circuit in frequency domain

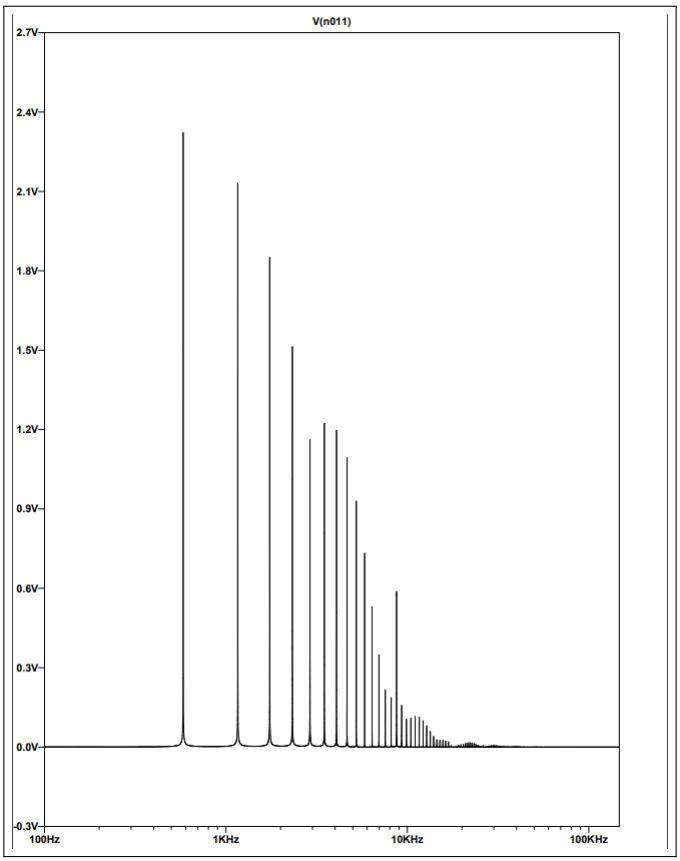

Figure 3.2.9: Linear: Inductor and capacitor circuit in frequency domain

iv) $R = 22\Omega$

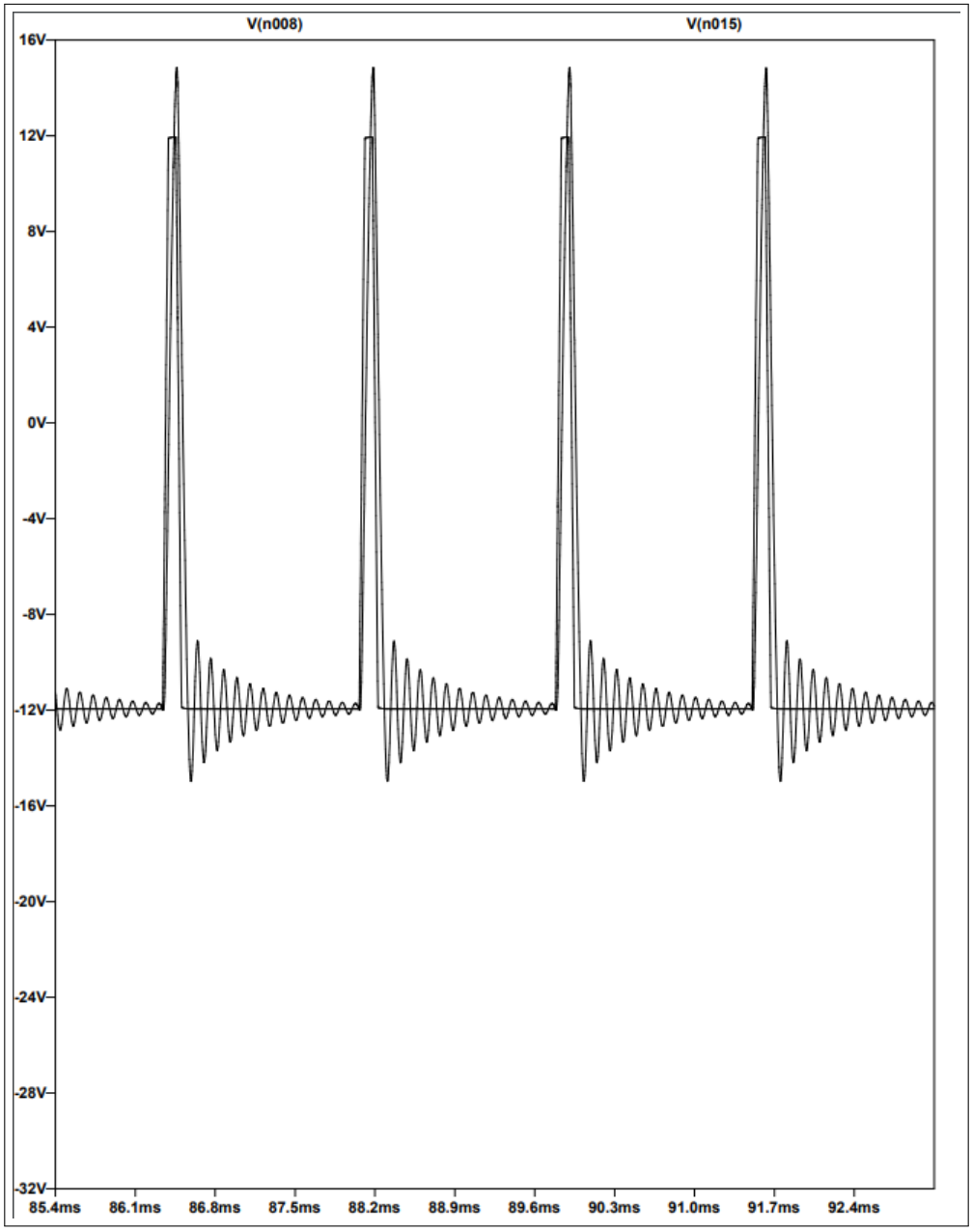

Figure 3.2.10: Signal generator in time domain with inductor and capacitor circuit

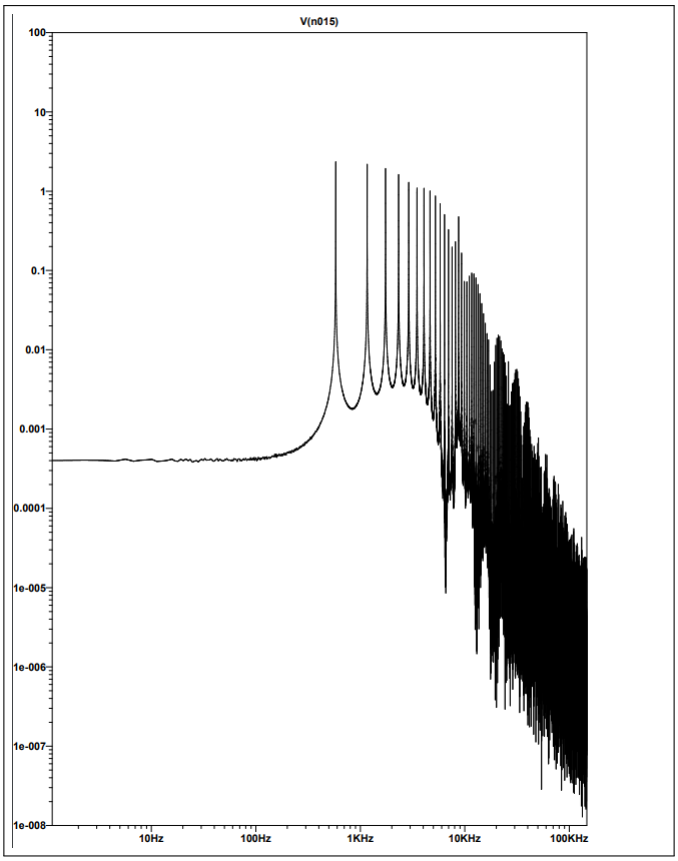

Figure 3.2.11: Logarithmic: Inductor and capacitor circuit in frequency domain

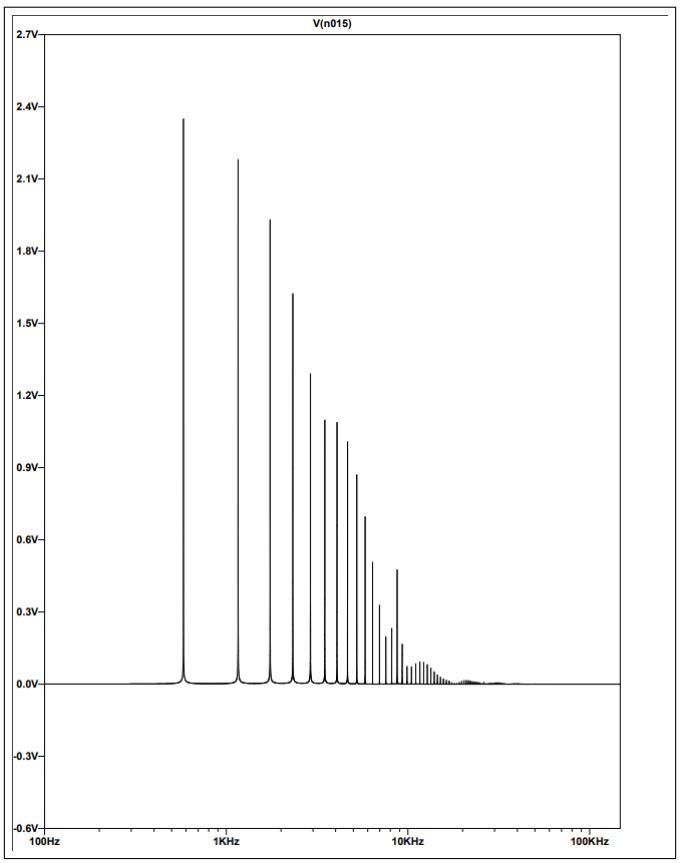

Figure 3.2.12: Linear: Inductor and capacitor circuit in frequency domain

v) $R = 680\Omega$

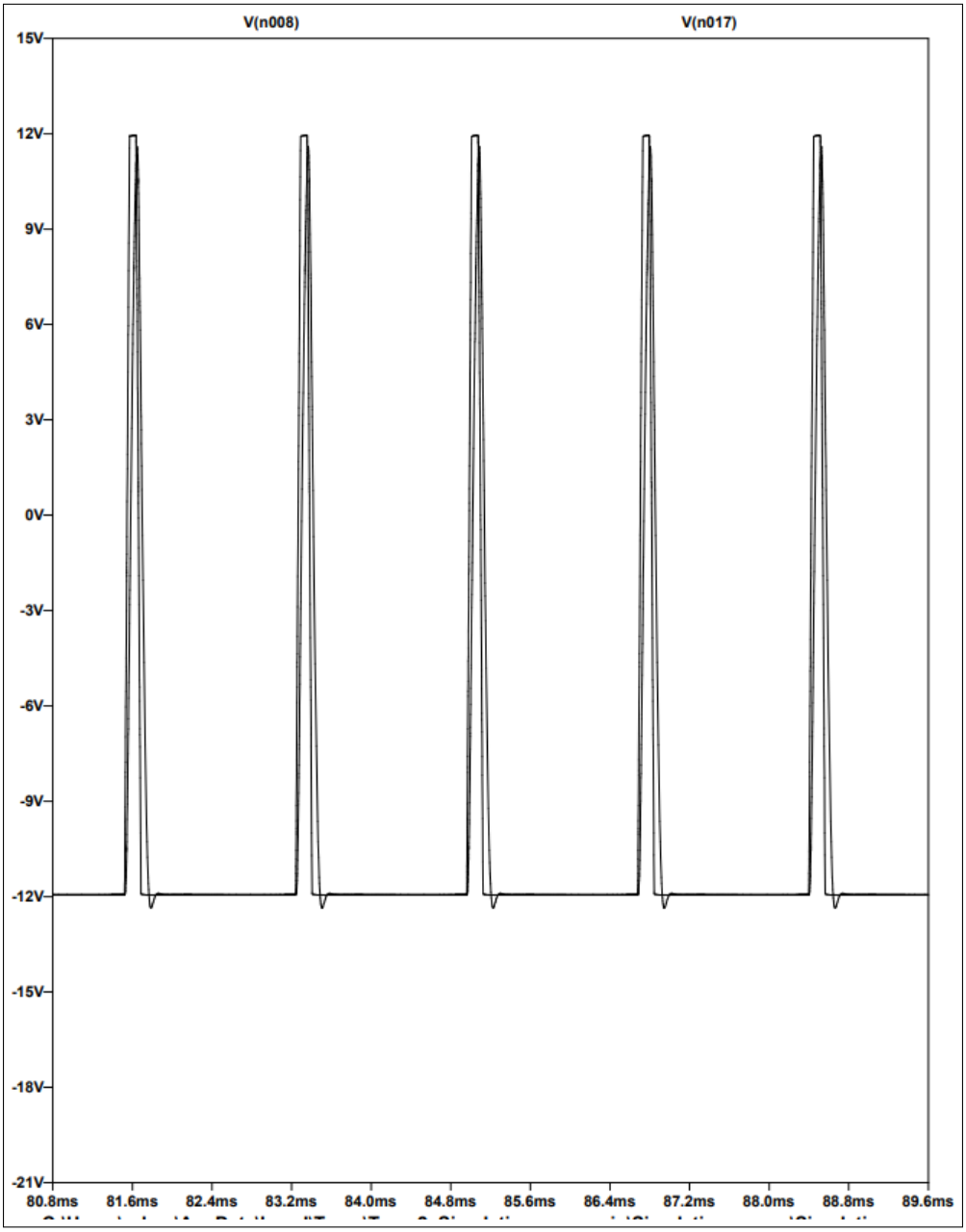

Figure 3.2.13: Signal generator in time domain with inductor and capacitor circuit

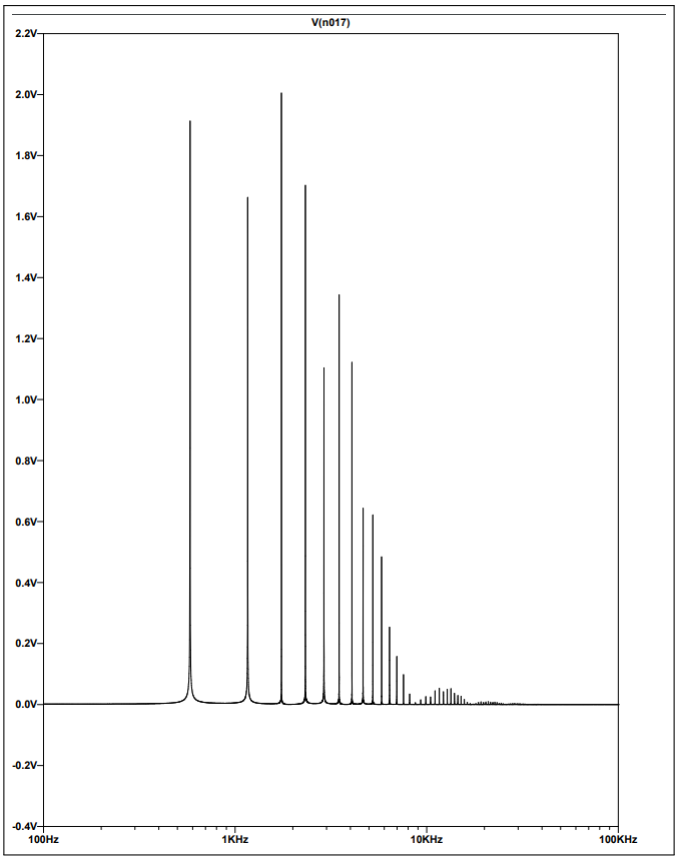

Figure 3.2.14: Logarithmic: Inductor and capacitor circuit in frequency domain

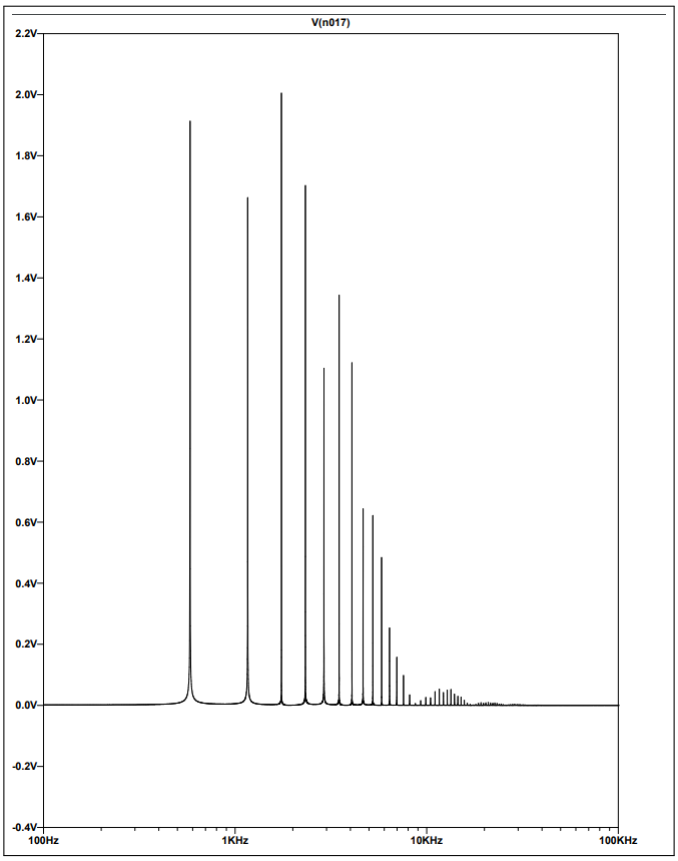

Figure 3.2.15: Linear: Inductor and capacitor circuit in frequency domain

vi) $R = 10k\Omega$

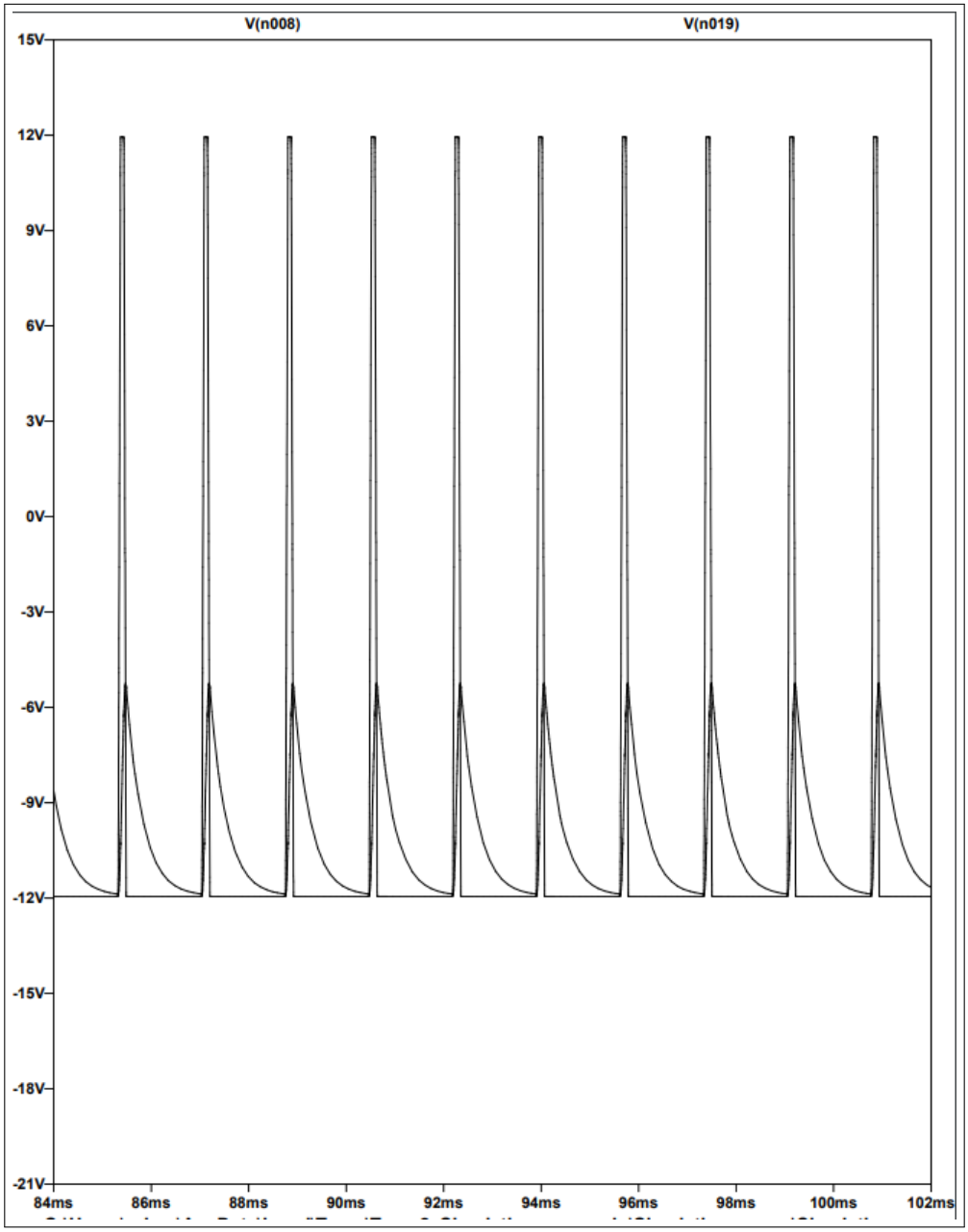

Figure 3.2.16: Signal generator in time domain with inductor and capacitor circuit

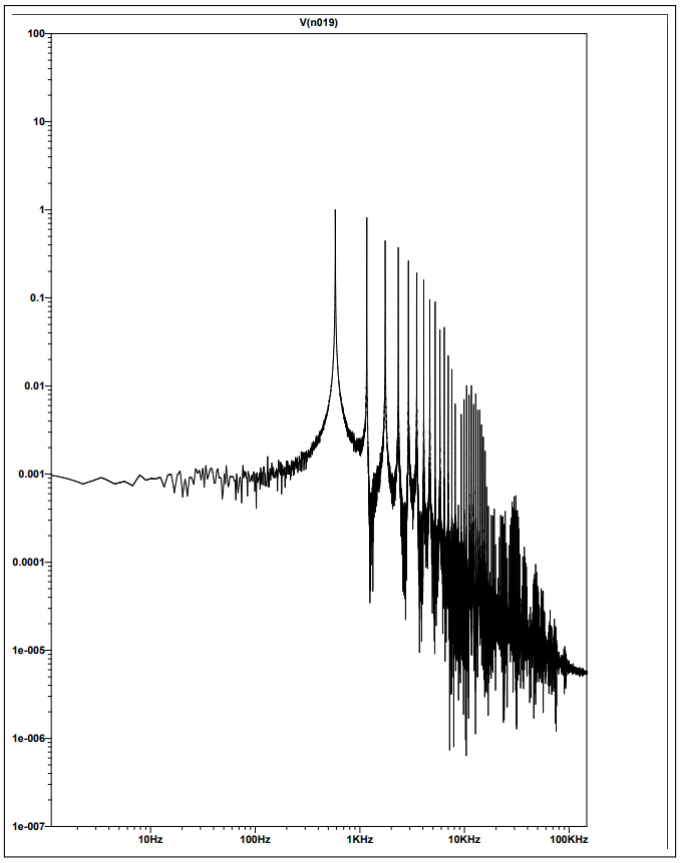

Figure 3.2.17: Logarithmic: Inductor and capacitor circuit in frequency domain

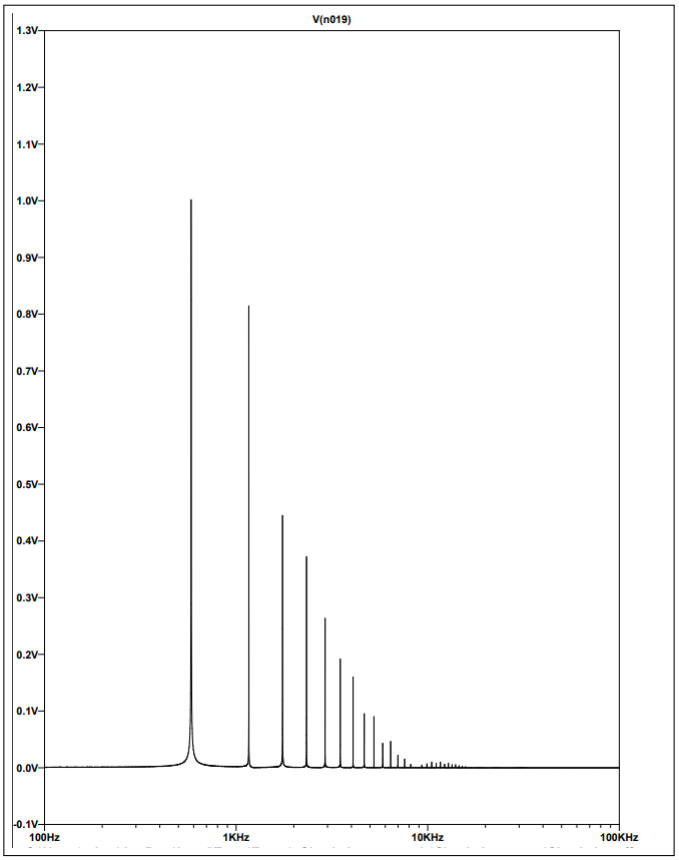

Figure 3.2.18: Linear: Inductor and capacitor circuit in frequency domain

4 Discussion

With a decrease of the value of the graphing lead to a decrease in the number of repetitions in the graph as this had an effect on the frequency of the graph. This decrease can be observed in the slow decrease in the amount of the peaks as the size of decreases.

The simulations of the RC, RL and RLC circuits where successful and the theoretical math supports the results that could be observed when the practical circuit was simulated. There was one difference that could be observed that was not as expected between the math and practical simulation that being question 2iii, where the signal generator has an effect on the output of the RLC circuit with no resistance this is not accounted for in the theoretical math but shows in the practical simulation.

The calculation of the math took the most time of all other aspects of the practical as the math was the basis of the simulations in whether it was as can be theoretically expected. As observed in the graphing with an increase in the frequency in the frequency domain there was an increase and then a decrease in the amplitude of the function H(t).

Overdamped, Underdamped and Critically damped all had effect on the time response of the circuit. As can be observed in the graphing of 2)a)i on page 7 and 2)a)i on page 30, it can be observed how the capacitor charges till it reaches a peak and the discharges over time. In the graphing of 2)a)ii on page 10 and 2)a)ii on page 32, we observe the quick changes in the change of voltage of the circuit this can be credited to the circuit being overdamped and the highs and lows that are observed.

In the RLC circuit in the time domain it can be observed that there is a loading of the capacitor which reaches a high peak and is then discharged overtime where an oscillation occurs, because of the inductor connected in the circuit. With an increase in the resistance these oscillations decrease as the resistor decreases the available current to the inductor. Overdamping of the circuit leads to a exponential decrease as can be see in the graphing of the simulations. Underdamping leads to a big rise in the voltage and then a oscillation over time.

When the resistor value of the signal generator was changed it led to a decrease in the pulse width of the signal generators output. This had an effect in the frequency domain of decreasing the pulse size that was observed. This effect is caused as the amount of power in the frequency domain is calculated by $10\log(P)$ and power in a line is calculated through $P = I^2R$ and with a dractic decrease of power with the decrease in resistance from 10kΩ to 1kΩ we can observe smaller pulses in the frequency domain.

5 Conclusion

In this practical the number of different pulse-signals were successfully simulated using LTspice and represented graphically using the appropriate modules in python. The simulation in figure 3.2.7 on page 34 was not graphically as expected, this could be due to the pulse train not being included in the calculations and thus the peak of a pulse train can not be observed in the theoretical graphing. All practical outcomes was thus achieved successfully.

Bibliography

[1] B. Lathi, *Linear systems andsignal*. Oxford university press, 2th ed., 2010.

Appendix A

Python Code

Prepatory Work Code
1 a)

```
import matplotlib.pyplot as plt
import numpy as np

def func(f):
    return np.pi*(np.sin(np.pi*f/2)/(np.pi*f/2))

x = np.linspace(-20, 20, 1000, endpoint=True)
y = [func(i) for i in x]

plt.ylabel("c(f)")
plt.xlabel("Frequency(Hz)")
plt.plot(x, y)
plt.grid(True)
plt.show()
```

1 b)

```
import matplotlib.pyplot as plt
import numpy as np

def func(f):
    return (1/20)*(np.sinc((2*np.pi*f*(1/20))/2))*2*np.pi

x = np.linspace(-40, 40, 500, endpoint=True)
y = [func(i) for i in x]

plt.ylabel("c(f)")
plt.xlabel("Frequency(Hz)")
plt.plot(x, y)
plt.grid(True)
plt.show()
```

1 c)

```
import matplotlib.pyplot as plt
import numpy as np

def func(f):
    pass

x = np.linspace(-30, 30, 1000, endpoint=True)
y = np.zeros(x.size)

plt.ylabel("c(f)")
plt.xlabel("Frequency(Hz)")
plt.plot(x, y)
plt.grid(True)
plt.show()
```

1 d)

```
import matplotlib.pyplot as plt
import numpy as np

def func(f):
    t = 75*pow(10, -6)
    T = 3.7*pow(10, -3)
    return t*(np.sinc((2*np.pi*f*t)/2))*((2*np.pi)/T)

x = np.linspace(-30, 30, 1000, endpoint=True)
y = [func(i*1000) for i in x]

plt.ylabel("c(f)")
plt.xlabel("Frequency(KHz)")
plt.plot(x, y)
plt.grid(True)
plt.show()
```

2 a)i)

```
import matplotlib.pyplot as plt
import numpy as np

def hf(f):
    return 1/(np.sqrt(pow(2*np.pi*f, 2)+pow(3030.30, 2)))

def ht(f):
    return np.exp(-3030.30*f)

xf = np.linspace(1, 1000000, 1000, endpoint=True)
```

```
xt = np.linspace(0, 1.5*pow(10, -3), 1000, endpoint=True)
yf = [hf(i) for i in xf]
yt = [ht(i) for i in xt]

fig, axs = plt.subplots(2)
axs[0].plot(xf, yf, label="|H(f)|")
axs[0].set_yscale("log")
axs[0].set_xscale("log")
axs[0].set(ylabel="|H(f)|", xlabel="Frequency(Hz)")
axs[1].set(ylabel="h(t)", xlabel="Time(s)")
axs[1].plot(xt, yt, label="h(t)")
axs[0].grid(True)
axs[0].legend()
axs[1].grid(True)
axs[1].legend()
plt.show()
```

2 a)ii)

```
import matplotlib.pyplot as plt
import numpy as np

def hf(f):
    return np.sqrt(1 + pow(5600/(pow(2*np.pi*f, 2)+5600), 2))

def ht(f):
    return -5600*np.exp(-5600*f)

xf = np.linspace(1, 100000000, 1000, endpoint=True)
xt = np.linspace(0, 2*pow(10, -3), 1000, endpoint=True)
yf = [hf(i) for i in xf]
yt = [ht(i) for i in xt]

fig, axs = plt.subplots(2)
axs[0].plot(xf, yf, label="|H(f)|")
axs[0].set_yscale("log")
axs[0].set_xscale("log")
axs[0].set(ylabel="|H(f)|", xlabel="Frequency(Hz)")
axs[1].set(ylabel="h(t)", xlabel="Time(s)")
axs[1].plot(xt, yt, label="h(t)")
axs[0].grid(True)
axs[0].legend()
axs[1].grid(True)
axs[1].legend()
plt.show()
```

2 a)iii)

```
import matplotlib.pyplot as plt
import numpy as np
```

```
def hf(f):
    return 1/(np.sqrt(pow(2*np.pi*f, 2)+pow(3030.30, 2)))

def ht(f):
    return -1.8165*pow(10, -5)*np.cos(55048.19*f + (np.pi/2))

xf = np.linspace(1, 1000000, 1000, endpoint=True)
xt = np.linspace(0, 4*pow(10, -3), 1000, endpoint=True)
yf = [hf(i) for i in xf]
yt = [ht(i) for i in xt]

fig, axs = plt.subplots(2)
axs[0].plot(xf, yf, label="|H(f)|")
axs[0].set_yscale("log")
axs[0].set_xscale("log")
axs[0].set(ylabel="|H(f)|", xlabel="Frequency(Hz)")
axs[1].set(ylabel="h(t)", xlabel="Time(s)")
axs[1].plot(xt, yt, label="h(t)")
axs[0].grid(True)
axs[0].legend()
axs[1].grid(True)
axs[1].legend()
plt.show()
```

2 a)iv)

```
import matplotlib.pyplot as plt
import numpy as np

def hf(f):
    return 1/(np.sqrt(pow(2*np.pi*f, 4)+pow(55048.19, 2)+pow(4400*np.pi*f, 2)))

def ht(f):
    return -1.8165*pow(10, -5)*np.exp(-1100*f)*np.cos(55037.19*f+(np.pi/5))

xf = np.linspace(1, 1000000, 1000, endpoint=True)
xt = np.linspace(0, 4*pow(10, -3), 1000, endpoint=True)
yf = [hf(i) for i in xf]
yt = [ht(i) for i in xt]

fig, axs = plt.subplots(2)
axs[0].plot(xf, yf, label="|H(f)|")
axs[0].set_yscale("log")
axs[0].set_xscale("log")
axs[0].set(ylabel="|H(f)|", xlabel="Frequency(Hz)")
axs[1].set(ylabel="h(t)", xlabel="Time(s)")
axs[1].plot(xt, yt, label="h(t)")
axs[0].grid(True)
axs[0].legend()
```

```
axs[1].grid(True)
axs[1].legend()
plt.show()
```

2 a)v)

```
import matplotlib.pyplot as plt
import numpy as np

def hf(f):
    return 1/(np.sqrt(pow(2*np.pi*f, 4)+pow(34000, 2)+pow(43293.22, 4)+pow(6800*2*np.pi*

def ht(f):
    return -2.309*pow(10, -5)*np.exp(-34000*f)*np.cos(43293.22*f+(np.pi/2))

xf = np.linspace(1, 1000000, 1000, endpoint=True)
xt = np.linspace(0, 300*pow(10, -6), 1000, endpoint=True)
yf = [hf(i) for i in xf]
yt = [ht(i) for i in xt]

fig, axs = plt.subplots(2)
axs[0].plot(xf, yf, label="|H(f)|")
axs[0].set_yscale("log")
axs[0].set_xscale("log")
axs[0].set(ylabel="|H(f)|", xlabel="Frequency(Hz)")
axs[1].set(ylabel="h(t)", xlabel="Time(s)")
axs[1].plot(xt, yt, label="h(t)")
axs[0].grid(True)
axs[0].legend()
axs[1].grid(True)
axs[1].legend()
plt.show()
```

2 a)vi)

```
import matplotlib.pyplot as plt
import numpy as np

def hf(f):
    return np.sqrt(pow((1.006*pow(10, -6))/(2*np.pi*f + 3039.54), 2) + pow((1.006*pow(10

def ht(f):
    return np.exp(-3030.30*f)

xf = np.linspace(1, 1000000, 1000, endpoint=True)
xt = np.linspace(0, 1.5*pow(10, -3), 1000, endpoint=True)
yf = [hf(i) for i in xf]
yt = [ht(i) for i in xt]
```

```
fig, axs = plt.subplots(2)
axs[0].plot(xf, yf, label="|H(f)|")
axs[0].set_yscale("log")
axs[0].set_xscale("log")
axs[0].set(ylabel="|H(f)|", xlabel="Frequency(Hz)")
axs[1].set(ylabel="h(t)", xlabel="Time(s)")
axs[1].plot(xt, yt, label="h(t)")
axs[0].grid(True)
axs[0].legend()
axs[1].grid(True)
axs[1].legend()
plt.show()
```